Title Page

SBT-DF204 - Investigating Memory Evidence Case Study Guide

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INTRODUCTION

Digital forensic investigations often involve examining information left behind on a computer to understand activities that may have taken place. One useful source of evidence is a computer's Random Access Memory (RAM), which can contain information about running applications, network connections, command history, and system activities.

This case study investigates a suspected unauthorized file transfer using a Windows memory image named memdumpWin7.mem. The main purpose of the investigation was to examine the available memory evidence and determine whether it supported the possibility that a document had been copied from the computer to an external storage device.

I carried out the investigation using Kali Linux and Volatility 2.6.1 in a VMware virtual environment. Before examining the memory image, I created a forensic working copy and verified its integrity using SHA-256 hashing. I then examined system processes, network connections, command history, Windows registry records, removable-storage information, and File Explorer artefacts.

During the examination, I recovered command history showing an instruction to copy a document named secret_file.docx from the Downloads folder to drive F:. The corresponding console output reported that one file had been copied. Further analysis of USB storage and registry records established that drive F: was associated with a General UDisk storage device.

The investigation also identified records associated with the IEUser Windows account, including ShellBag information and process security identifiers. These findings were examined together to establish an evidence-based account of the suspected activity.

Throughout the investigation, I distinguished information directly recovered from the memory image from conclusions that could only be inferred. This approach was important because evidence of a file-copy operation does not automatically establish that the activity was unauthorized or identify the person responsible.

OBJECTIVES

The main objective of this investigation was to examine the supplied Windows memory image and determine whether the recovered digital evidence supported a suspected unauthorized file transfer.

The specific objectives were to:

1. Identify and preserve the original memory image, create a forensic working copy, and verify its integrity using SHA-256 hashing.
2. Determine the operating system, memory profile, and recorded acquisition time using Volatility.
3. Examine running processes and process relationships to identify applications potentially connected with the suspected activity.
4. Investigate recovered network connections and command history to identify possible file-transfer activity.
5. Examine Windows registry records to identify USB storage devices and determine whether drive F: was associated with removable storage.
6. Analyse user-profile information, security identifiers, and ShellBag records to establish relationships between the Windows account, folders, and mounted drives.
7. Reconstruct a timeline using the available timestamps while identifying inconsistencies and limitations in the recovered evidence.
8. Present findings supported by forensic evidence and distinguish documented computer activity from assumptions about authorization or individual responsibility.

Laboratory / Tools Used

The investigation was conducted in a virtual forensic laboratory using the following tools and utilities.

1. VMware Workstation

VMware Workstation provided the virtual environment used to run Kali Linux during the investigation. It allowed the forensic examination to be carried out within a dedicated laboratory environment.

2. Kali Linux

Kali Linux was the main operating system used throughout the examination. Its terminal provided the environment for organizing the evidence files, executing forensic commands, examining results, and maintaining investigation outputs.

3. Python 2.7.18

Python 2.7.18 was used to execute Volatility 2. The Python environment was checked before the memory analysis, and the necessary dependencies were examined during the tool setup.

4. Volatility Framework 2.6.1

Volatility 2.6.1 was the primary memory-analysis tool used in this investigation.

The following plugins were used:

imageinfo: Identified the operating-system characteristics, possible memory profiles, and image timestamps.

pslist and psscan: Examined active processes and process records recovered from memory.

pstree: Identified relationships between parent and child processes.

hivelist: Located Windows registry hives in the memory image.

netscan: Examined recorded network endpoints and connection information.

cmdscan and consoles: Recovered Command Prompt history and available console output.

printkey: Examined registry records relating to USB devices, mounted volumes, user profiles, and time-zone settings.

shellbags: Recovered Windows File Explorer folder-history information.

getsids: Identified Windows account security identifiers associated with selected processes.

5. Linux Terminal Utilities

Several command-line utilities were used to prepare and verify the evidence.

The mkdir command was used to create investigation folders, while ls, pwd, and stat were used to identify file locations and sizes.

The cp command created the working copy, and chmod was used to apply read-only permissions to the original memory image.

The sha256sum command calculated the memory images' SHA-256 hashes, while cmp confirmed that the original and working copies were identical.

Commands such as cat, grep, head, tail, less, and file were used to examine saved outputs and verify supporting files.

6. Memory Evidence

The evidence examined was a Windows memory image named memdumpWin7.mem, with a recorded size of 1,073,676,288 bytes.

The original image was preserved in the investigation directory, while a separate copy was used for the Volatility examination.

All forensic conclusions were based on the recovered information from this memory image and its associated analysis outputs.

Lab Folder Structure and Evidence Preparation

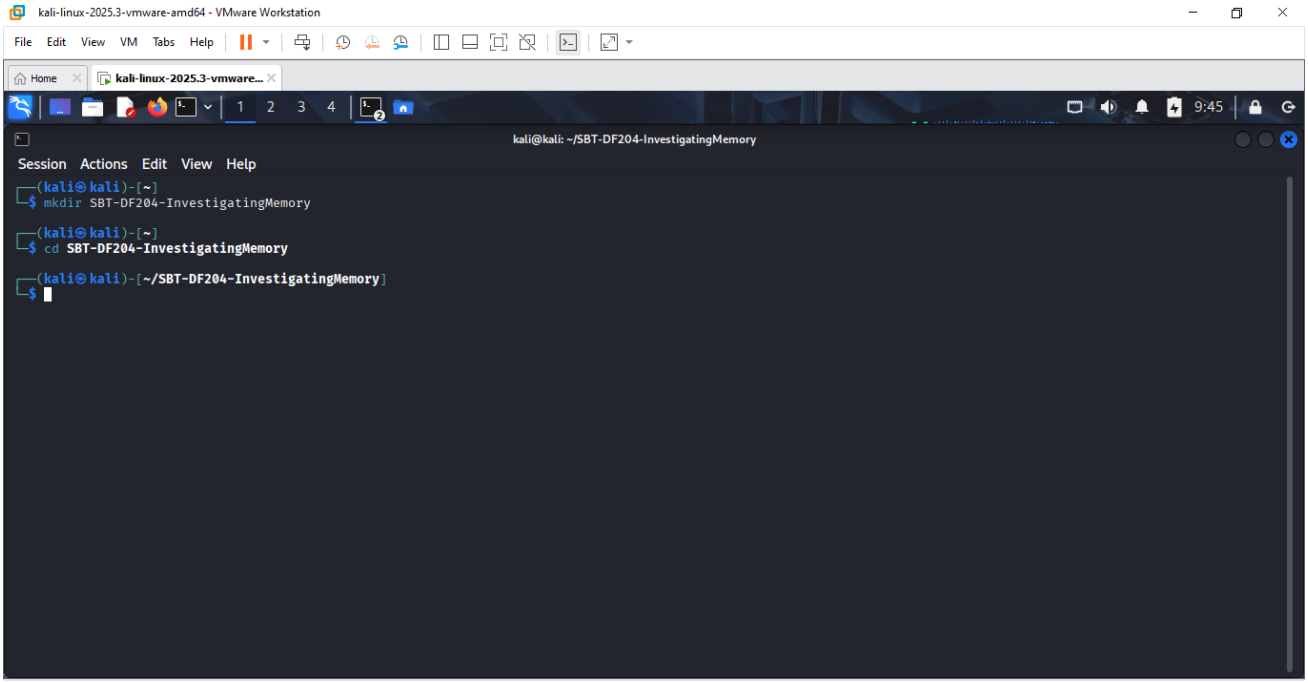


Fig 1.0 Creating the Lab folder.

Explanation

I created a dedicated investigation folder named SBT-DF204-InvestigatingMemory in Kali Linux to organize the examination. This provided a central location for the memory evidence, saved Volatility results, and supporting investigation files.

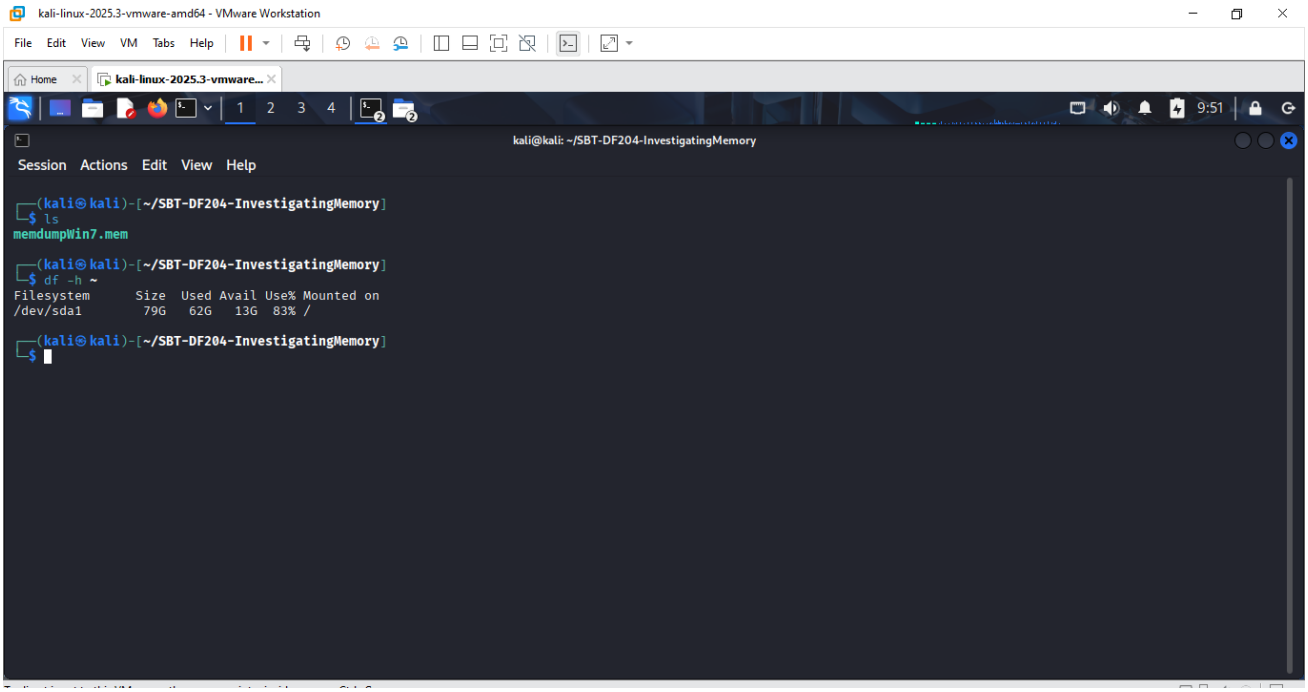


Fig 1.1 Original Memory Evidence Identification.

Explanation

I used the Kali Linux terminal to confirm that memdumpWin7.mem was available in the investigation directory. I also checked the available disk space to ensure the system could accommodate a separate forensic working copy without affecting the original file.

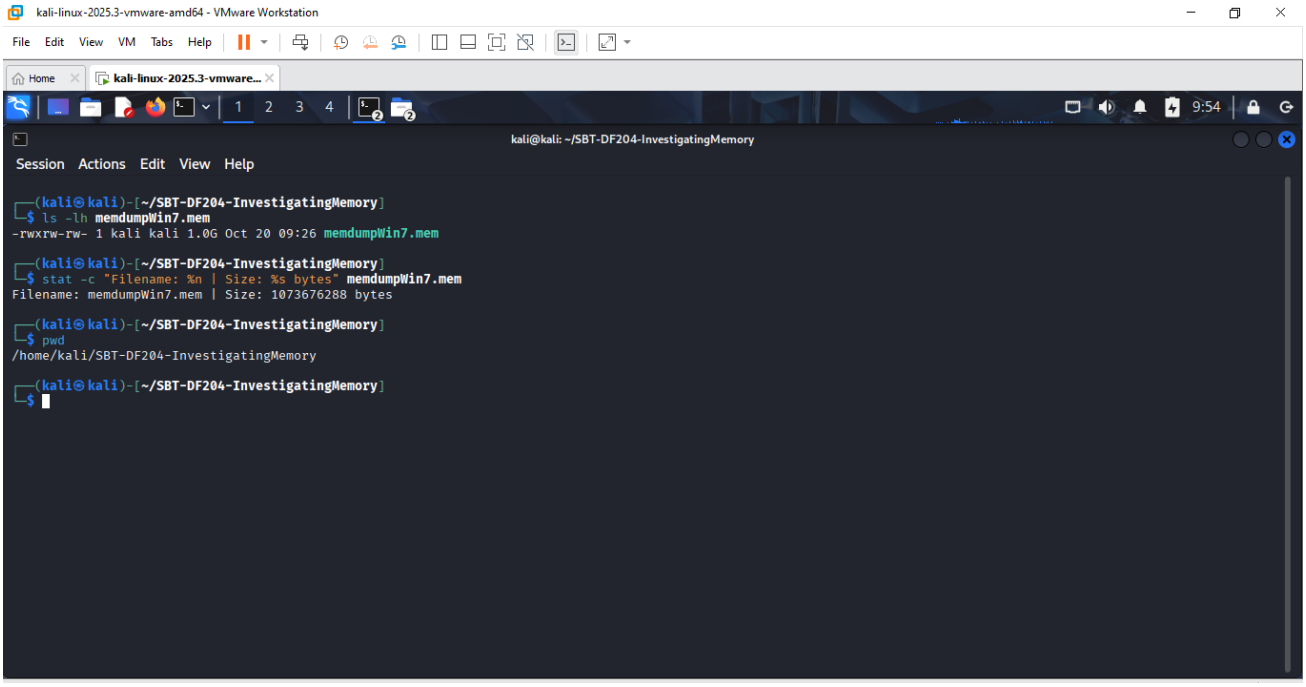


Fig 1.2 Original Memory Evidence Identification page 1.

Explanation

The commands displayed the memory image's filename, storage location, and exact size of 1,073,676,288 bytes. Recording these details established the identity of the evidence being examined and provided information needed for the acquisition and integrity section of the report.

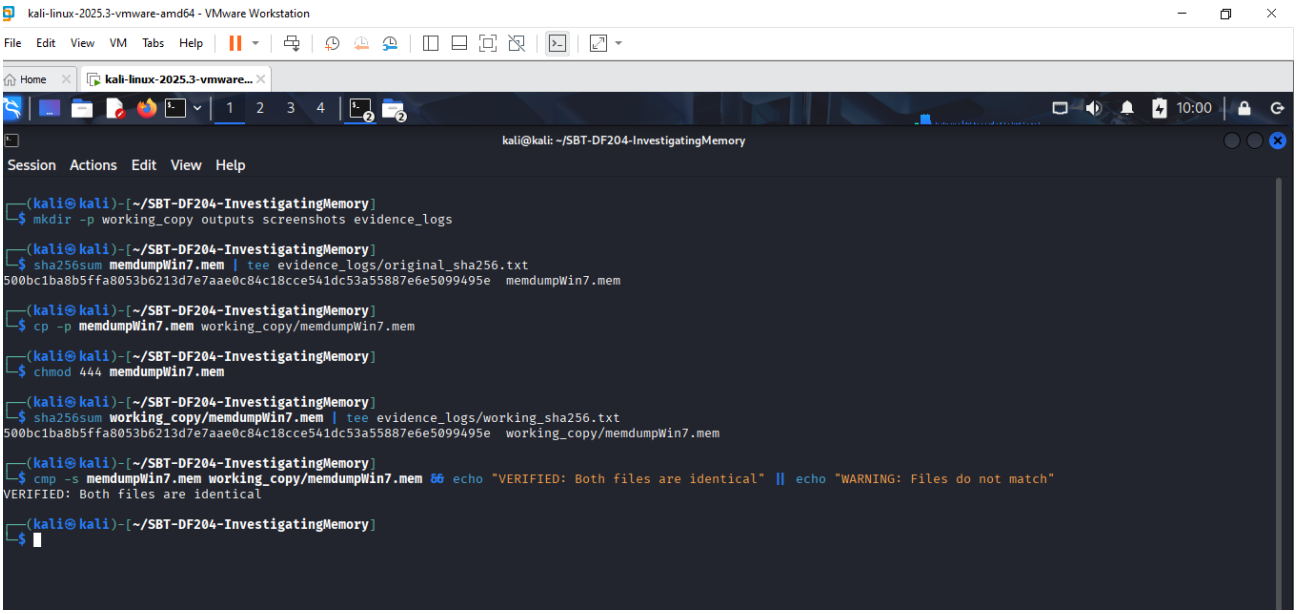


Fig 1.3 SHA256 Evidence Integrity Verification.

Explanation

I calculated the SHA-256 hashes of the original memory image and its working copy, and both files produced identical values. A further byte-for-byte comparison confirmed that the copies matched, allowing the investigation to proceed using the working image while preserving the original evidence.

Evidence Log — E01

Field	Details
Evidence ID	E01
Source Record	memdumpWin7.mem
Evidence Size	1,073,676,288 bytes
Original Location	/home/kali/SBT-DF204-InvestigatingMemory/
Working Copy	working_copy/memdumpWin7.mem
Finding	Both files produced identical SHA-256 hashes. Byte-for-byte comparison also confirmed that the files match.
Why It Matters	Establishes the integrity of the forensic working copy and allows subsequent analysis without modifying the original image.
Screenshot Reference	Fig 1.3 SHA256 Evidence Integrity Verification.

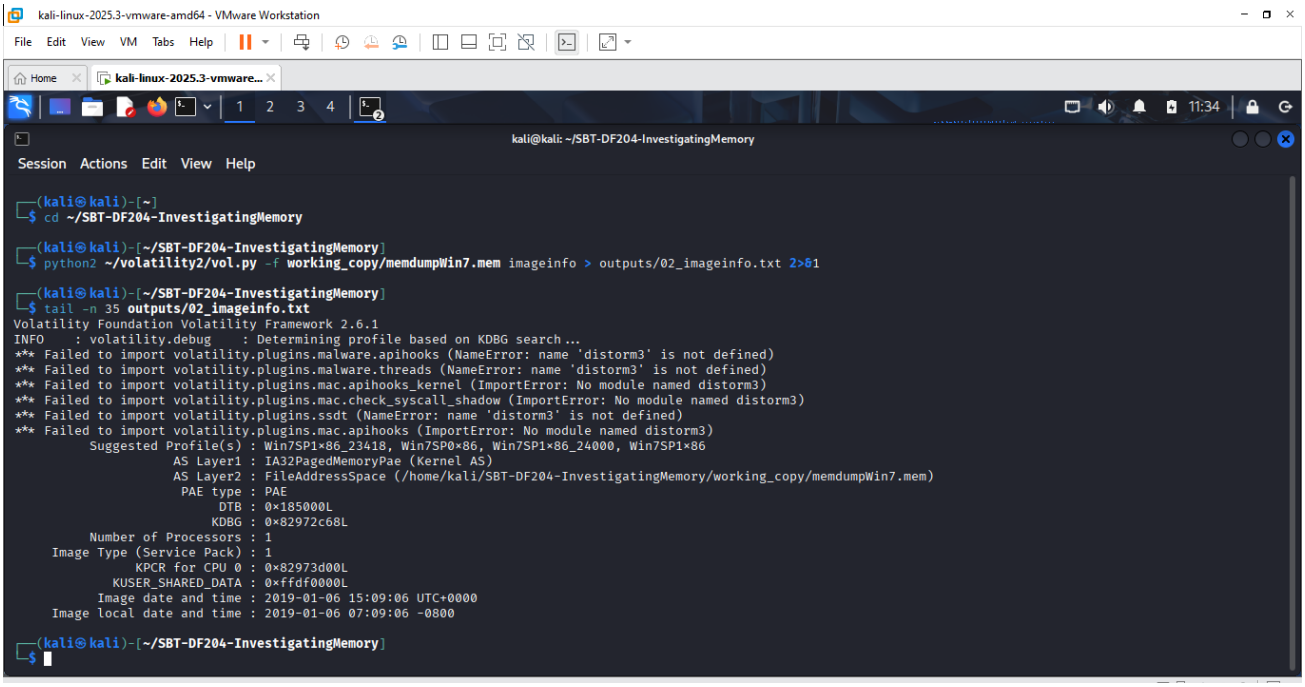


Fig 1.4 Memory Imageinfo Profile Identification.

Explanation

The Volatility imageinfo plugin examined the memory image and identified several possible Windows

profiles, with Win7SP1x86_23418 selected for further validation. It also identified the image timestamp as 6 January 2019 at 15:09:06 UTC and reported a local time eight hours behind UTC.

Memory Image Findings

Evidence Detail	Result
Forensic Tool	Volatility Framework 2.6.1
Memory Image	memdumpWin7.mem
Operating System	Windows 7 family
Architecture	32-bit (x86)
Service Pack	1
Number of Processors	1
Memory Configuration	PAE
DTB	0x185000
KDBG	0x82972c68
Image Date (UTC)	2019-01-06 15:09:06
Image Local Time	2019-01-06 07:09:06
Local Time Offset	UTC−08:00

Evidence Log — E02

Field	**Description**
Evidence ID	E02
Source Record	Volatility 2 imageinfo output
Finding	Memory analysis identified Windows 7 x86, Service Pack 1, with an image timestamp of 6 January 2019 at 15:09:06 UTC. Four candidate profiles were returned.
Why It Matters	Establishes the operating-system characteristics, provides the memory-image timestamp, and identifies candidate profiles for further validation.
Volatility Command / Screenshot	imageinfo — Fig 1.4 Memory Imageinfo Profile Identification.

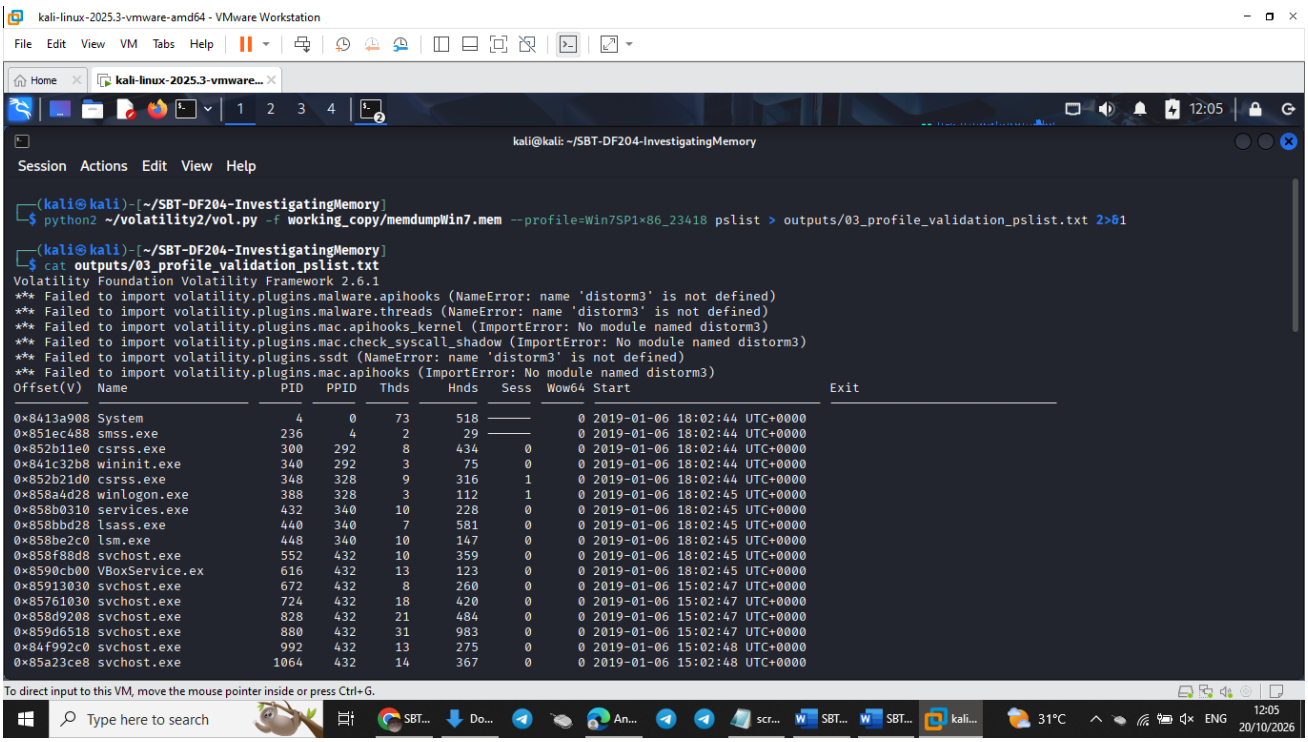


Fig 1.5 Memory Profile Validation Pslist.

Explanation

I used the pslist plugin with the selected memory profile to examine processes recorded in the Windows memory image. The results included core operating-system processes such as System, smss.exe, and services.exe, providing useful information for checking whether the chosen profile could interpret the memory correctly.

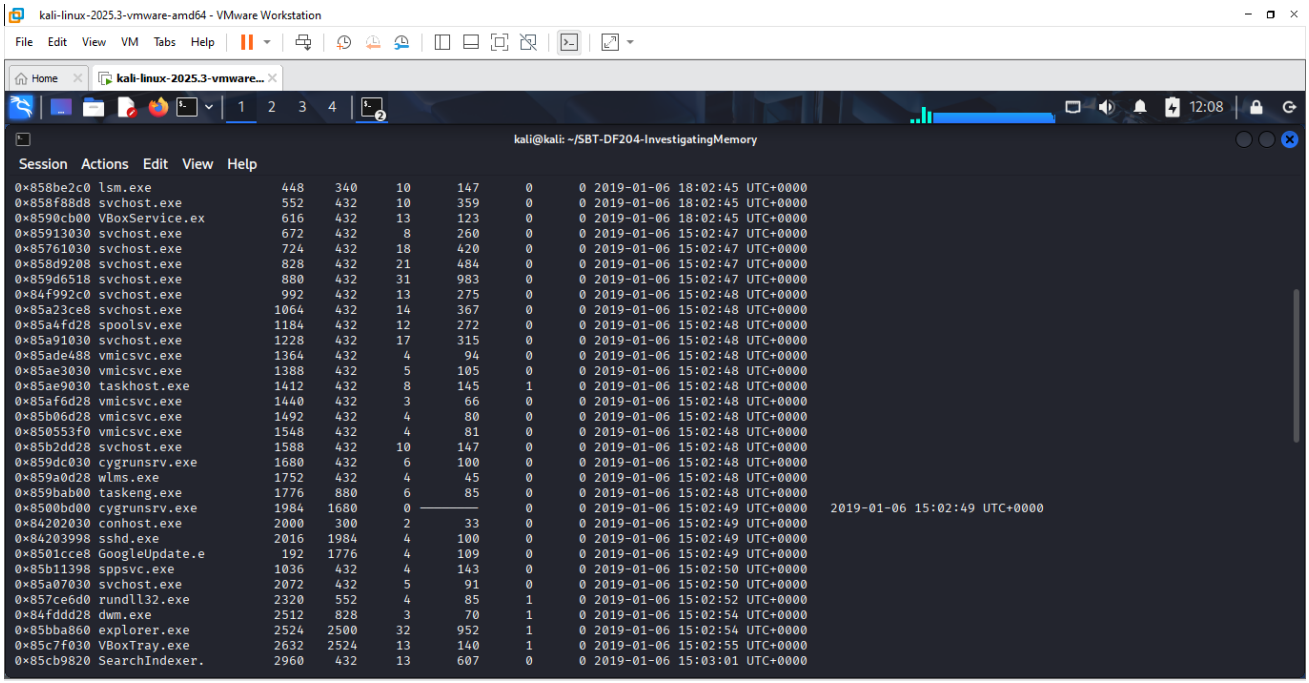


Fig 1.6 Memory Profile Validation Pslist page 1.

Explanation

This continuation of the process listing displayed additional Windows processes alongside their process IDs, parent process IDs, and creation timestamps. These records helped establish the range of system activity present in memory and provided information for later process correlation.

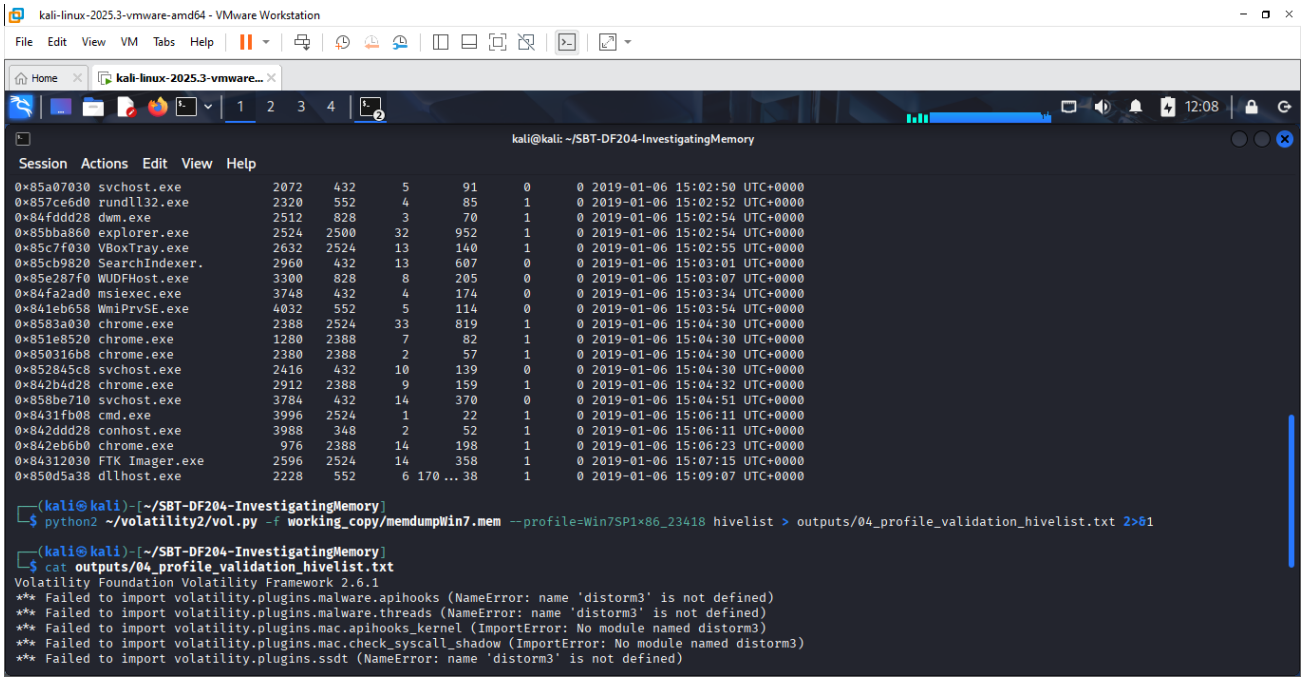


Fig 1.7 Memory Profile Validation Pslist page 2.

Explanation

The remaining process records included Windows Explorer, Google Chrome, Command Prompt, and FTK Imager. These applications were relevant because they could be examined further to understand browser activity, command execution, and possible file-handling operations.

Evidence Log — E03

Field	Description
Evidence ID	E03
Source Record	Volatility pslist
Finding	Identified active Windows processes, including Chrome, Command Prompt, Windows Explorer, and FTK Imager, with their PIDs, parent PIDs, and creation timestamps.
Why It Matters	Establishes process activity that can be correlated with command history, network records, and other memory artefacts.
Screenshot Reference	Fig 1.6 Memory Profile Validation Pslist

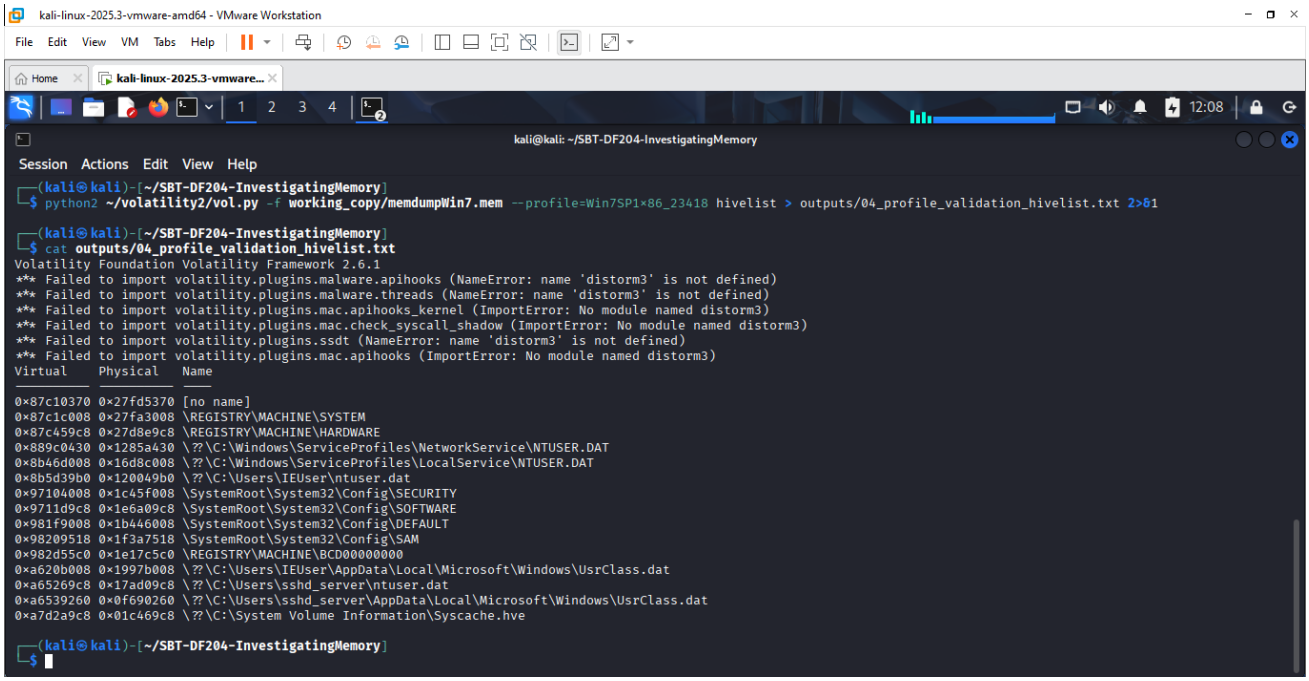


Fig 1.8 Memory Profile Validation Hivelist.

Explanation

The hivelist plugin identified Windows registry hives available within the memory image, including SYSTEM, SOFTWARE, SAM, and user-specific registry files. The recovered hive paths provided the foundation for examining Windows accounts, removable-storage records, drive assignments, and user-profile activity.

Evidence Log — E04

Field	Description
Evidence ID	E04
Source Record	Volatility hivelist
Finding	Identified SYSTEM, SOFTWARE, SAM, and user-profile registry hives, including records associated with IEUser and sshd_server.
Why It Matters	Provides registry-hive locations for investigating Windows accounts, USB devices, drive mappings, and user-profile activity.
Screenshot Reference	Memory Profile Validation Hivelist.

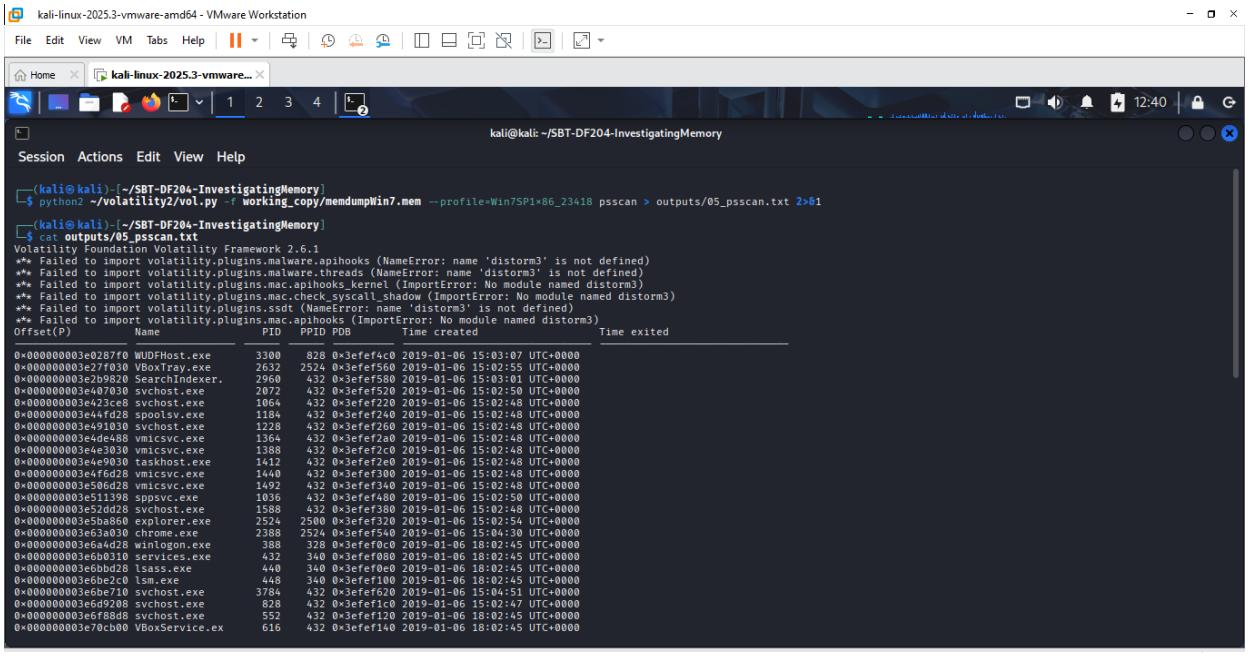


Fig 1.9 PSSCAN Process Investigation.

Explanation

I ran the psscan plugin to search memory for Windows process records, including objects that might not appear in the normal active-process listing. The recovered entries provided additional process information that could be compared with the earlier pslist results.

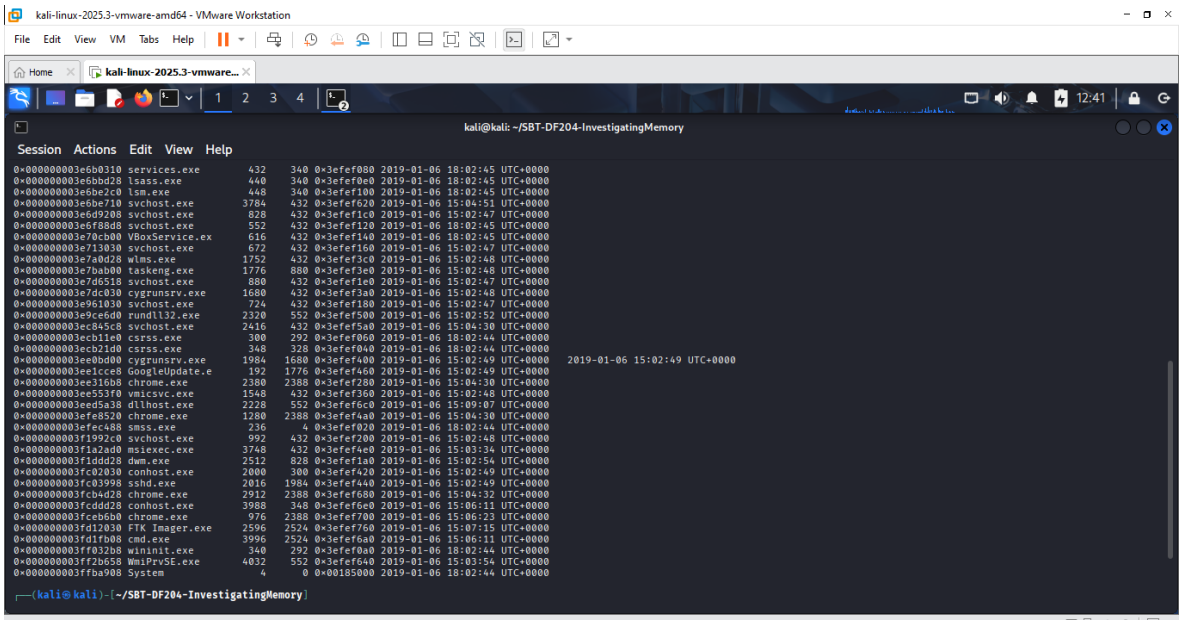


Fig 2.0 PSSCAN Process Investigation page 1.

Explanation

The continuation of the process scan displayed records associated with Chrome, Windows Explorer, Command Prompt, and other Windows applications. Their process identifiers and recorded timestamps

helped identify which applications required closer examination during the investigation.

E05 — Process Scan Investigation

Field	Description
Evidence ID	E05
Source Record	Volatility 2.6.1, psscan, profile Win7SP1x86_23418
Finding	Process records include Chrome, Windows Explorer, Command Prompt, and FTK Imager. Their PIDs, parent PIDs, and creation timestamps were recovered.
Why It Matters	Identifies processes and timestamps that can be correlated with network connections, command history, and other forensic artefacts.
Screenshot Reference	Fig 2.0 PSSCAN Process Investigation page 1.

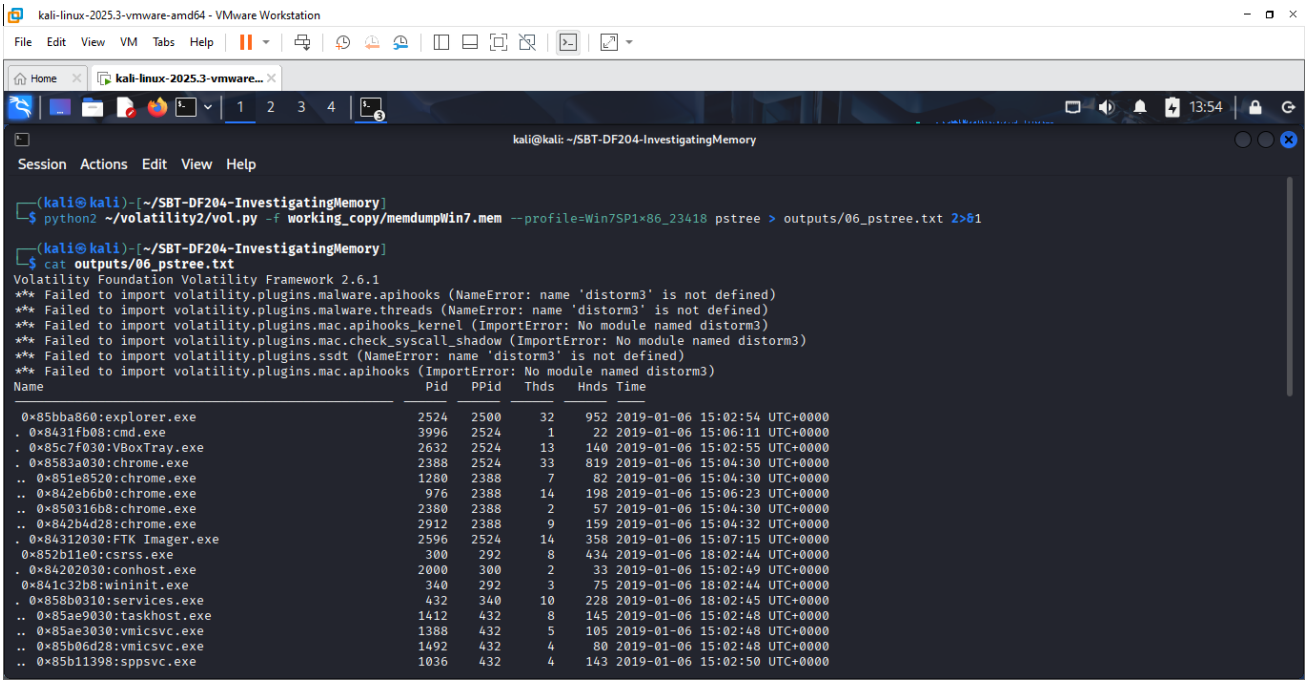


Fig 2.1 PSTREE Process Relationships.

Explanation

The pstree plugin displayed how the recovered processes were related to one another. Windows Explorer, PID 2524, appeared as the parent of Command Prompt, Chrome, and FTK Imager, helping establish the process hierarchy associated with the examined workstation.

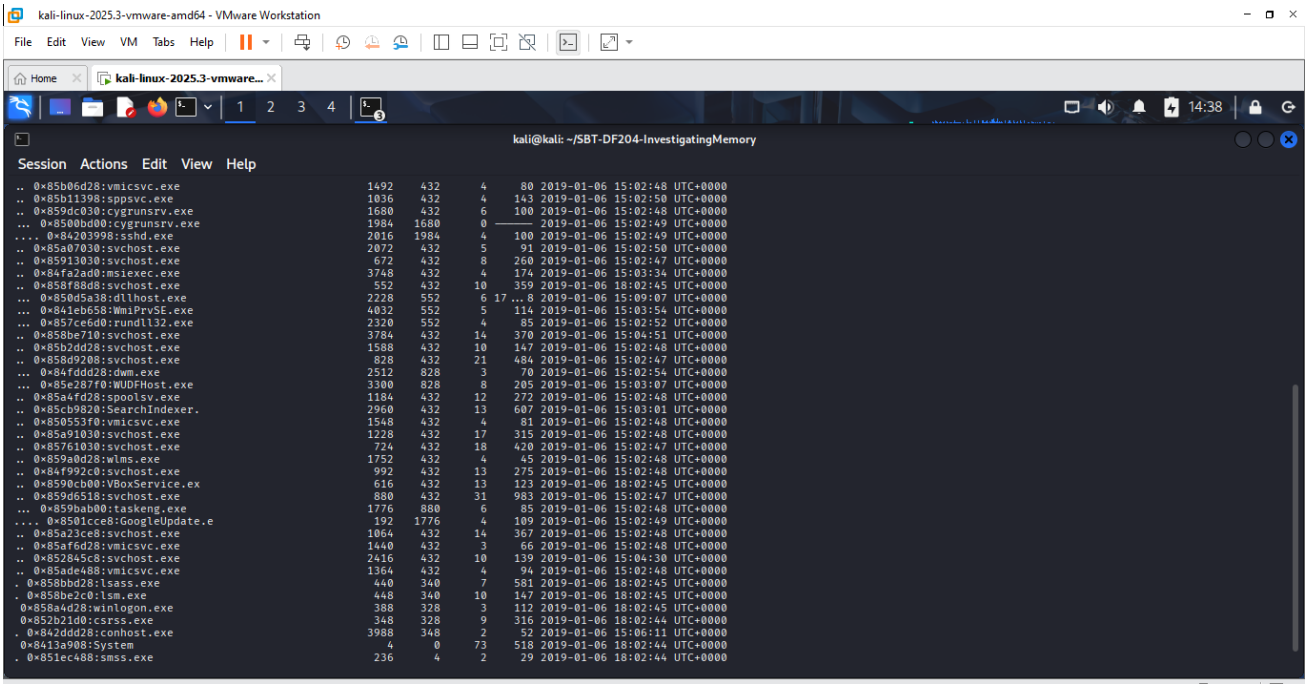


Fig 2.2 PSTREE Process Relationships page 1.

Explanation
The remaining process tree identified additional system processes and an SSH server process, sshd.exe, operating through Cygwin service processes. This finding was relevant to the network investigation because SSH can support remote communication, although the process relationship alone did not establish that a file transfer occurred.

E06 — Process Relationship Investigation

Field	Description
Evidence ID	E06
Source Record	Volatility 2.6.1 — pstree
Finding	The process hierarchy identified relationships involving Windows Explorer, Chrome, Command Prompt, FTK Imager, and an SSH server process (sshd.exe, PID 2016) launched through Cygwin service processes.
Why It Matters	Establishes process relationships relevant to investigating possible browser activity, command execution, and remote access. These records require correlation with network and command-history evidence before drawing conclusions about a transfer.
Screenshot Reference	Fig 2.1 PSTREE Process Relationships

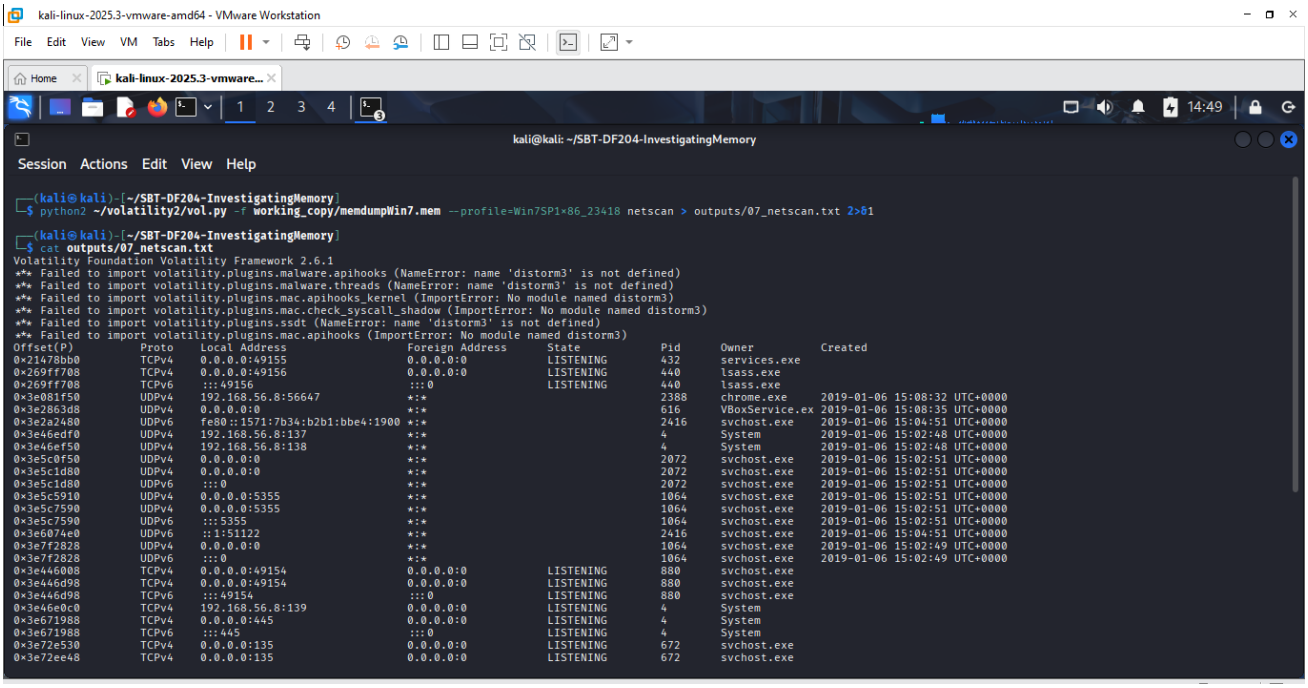


Fig 2.3 NETSCAN Network Connections.

Explanation

I used the Volatility netscan plugin to examine network information recovered from memory. The results identified local network endpoints, process associations, and listening services that could be compared with the processes already discovered during the investigation.

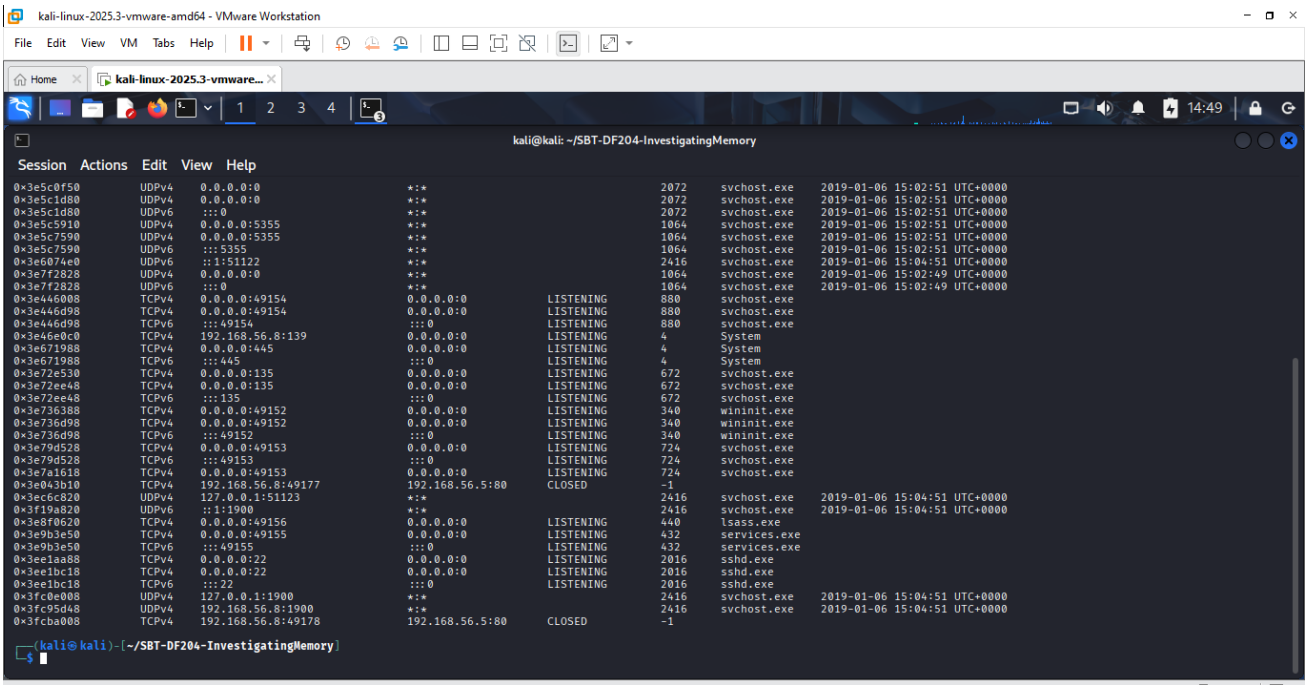


Fig 2.4 NETSCAN Network Connections page 1.

Explanation

The continuation of the network results showed two closed TCP connections involving 192.168.56.8 and

192.168.56.5 on port 80, as well as an SSH server listening on port 22. Although these records established network-related activity, they did not independently demonstrate that a document had been transferred across the network.

E07 — Network Connection Investigation

Field	Description
Evidence ID	E07
Source Record	Volatility 2.6.1 — netscan, profile Win7SP1x86_23418; outputs/07_netscan.txt
Finding	Identified a Chrome-associated UDP endpoint, two closed TCP connections to 192.168.56.5:80, and an SSH server listening on TCP port 22.
Why It Matters	Establishes network artefacts that can be correlated with browser activity, command history, and possible remote access. The records do not independently prove a completed file transfer.
Screenshot Reference	Fig 2.3 NETSCAN Network Connections.

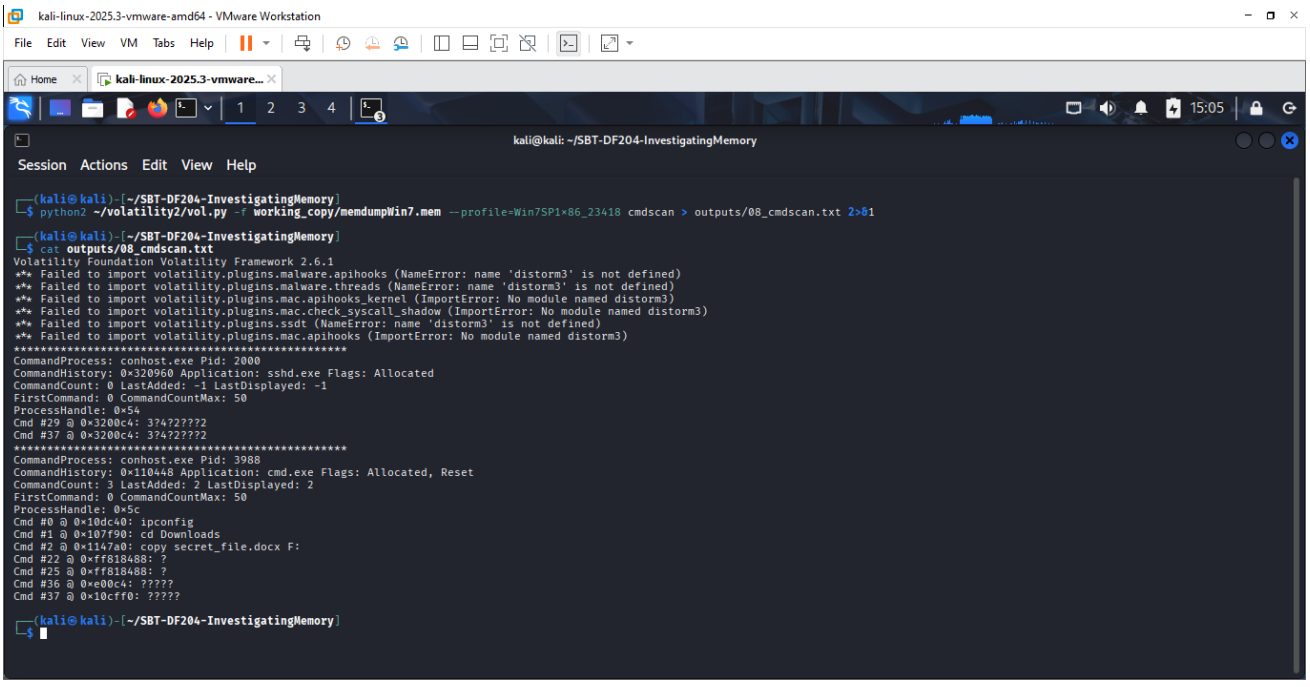


Fig 2.5 CMDSCAN Command History.

Explanation

The cmdscan plugin recovered commands associated with Command Prompt activity, including ipconfig, cd Downloads, and copy secret_file.docx F:. The copy command was particularly important because it identified a document in the Downloads directory and a destination drive relevant to the suspected transfer.

E08 — Command History Investigation

Field	Description
Evidence ID	E08
Source Record	Volatility 2.6.1 — cmdscan; cmd.exe PID 3996, conhost.exe PID 3988
Finding	Recovered ipconfig, cd Downloads, and copy secret_file.docx F: commands.
Why It Matters	Establishes that a command was issued to copy a document from the Downloads directory to drive F:.
Command / Screenshot Reference	cmdscan, profile Win7SP1x86_23418; Fig 2.5 CMDSCAN Command History.

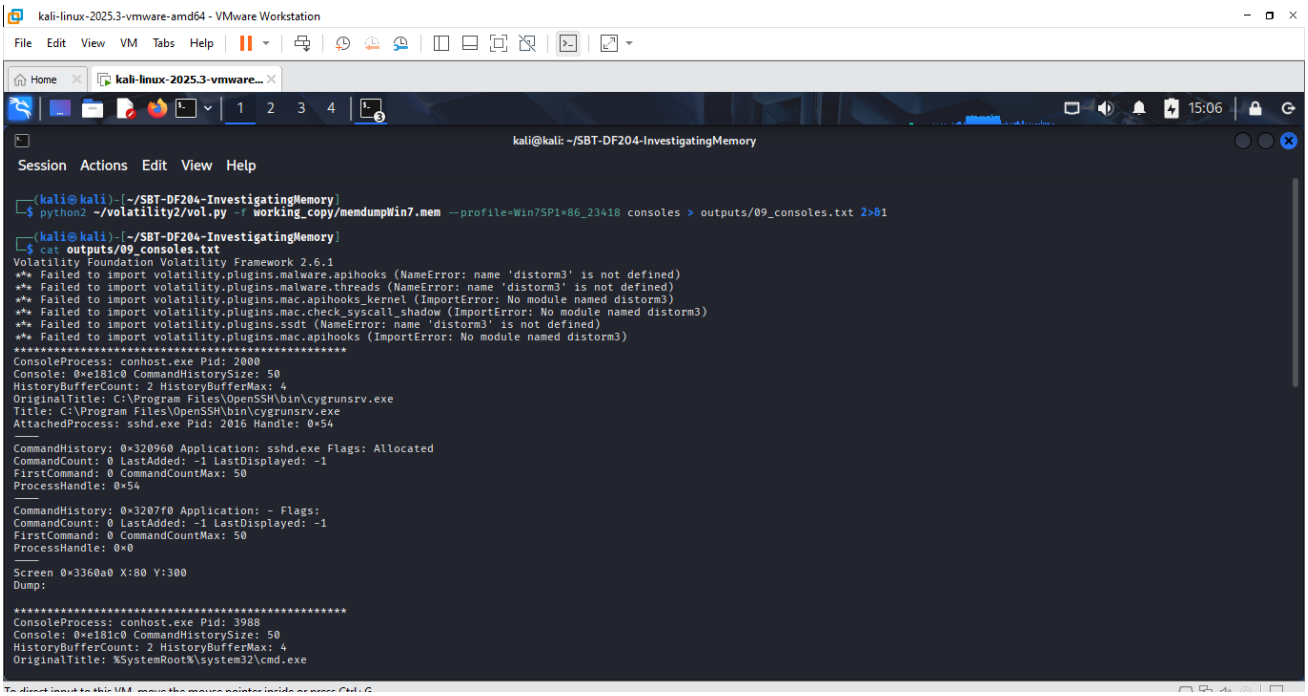


Fig 2.6 CONSOLES Command Execution Evidence.

Explanation

The consoles plugin recovered console-related information associated with the examined system, including command-history records. This examination was important because console buffers could provide additional information about what appeared after a command was entered.

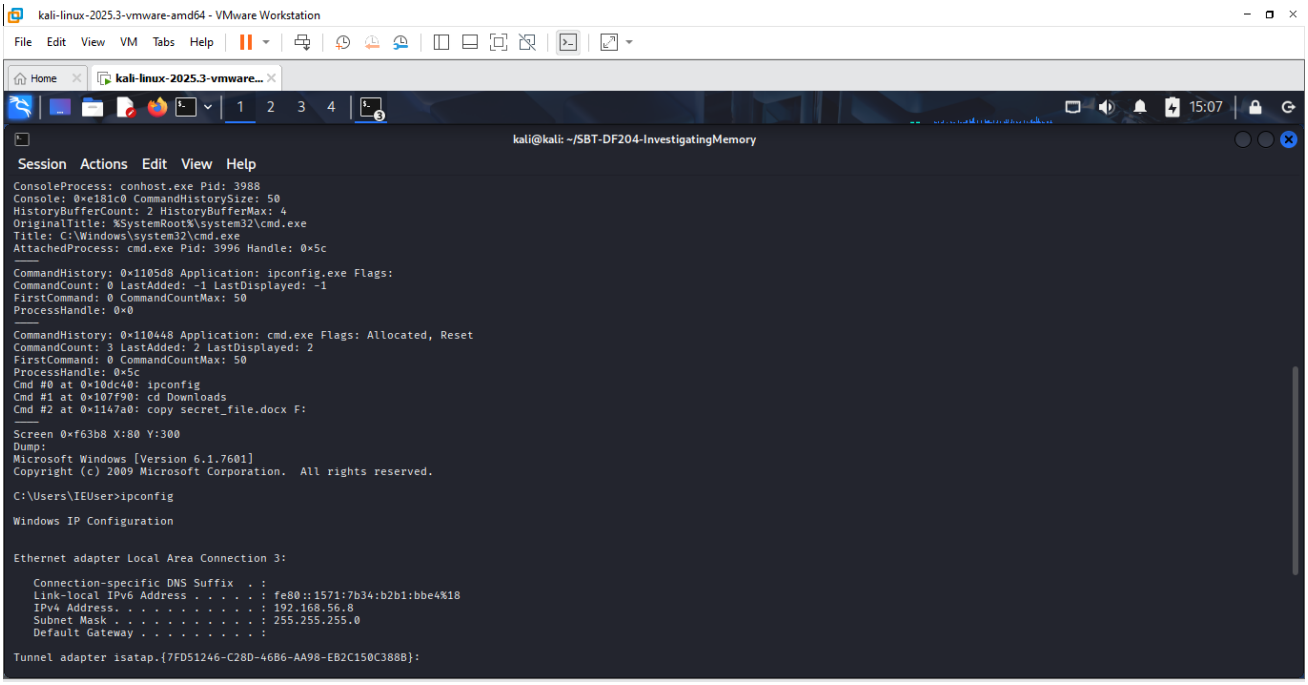


Fig 2.7 CONSOLES Command Execution Evidence page 1.

Explanation

The recovered console information displayed Windows network configuration details and command activity associated with the IEUser account. The workstation's IPv4 address, 192.168.56.8, was consistent with the local address identified during the earlier network examination.

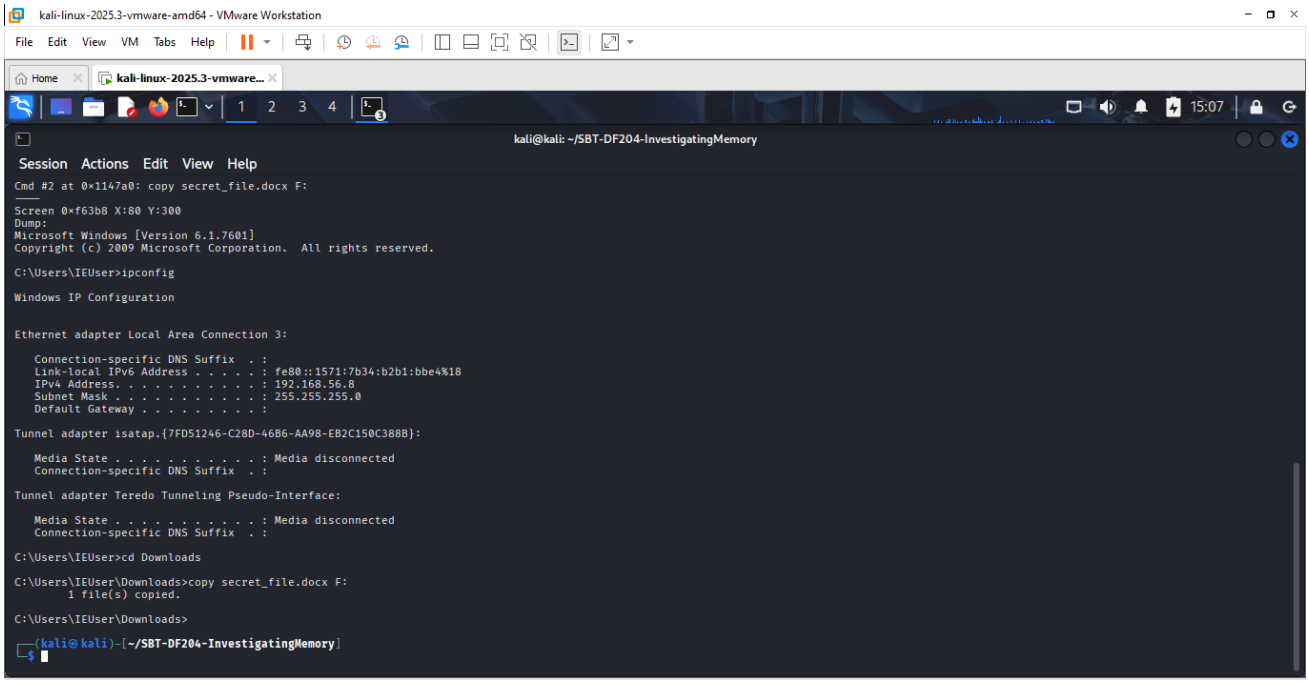


Fig 2.8 CONSOLES Command Execution Evidence page 2.

Explanation

This screenshot provided the clearest evidence relating to the suspected file transfer, showing copy secret_file.docx F: followed by 1 file(s) copied.. The recovered output indicates that Windows reported a

successful copy operation, although the destination file was not independently recovered or examined.

E09 — Console Output and Copy Confirmation

Field	Description
Evidence ID	E09
Source Record	Volatility 2.6.1 — consoles; cmd.exe PID 3996
Finding	Console output shows copy secret_file.docx F: followed by 1 file(s) copied.
Why It Matters	Corroborates that Windows reported successful execution of the file-copy operation. The destination device and copied contents still require verification.
Command / Screenshot Reference	consoles, profile Win7SP1x86_23418; CONSOLES Command Execution Evidence page.

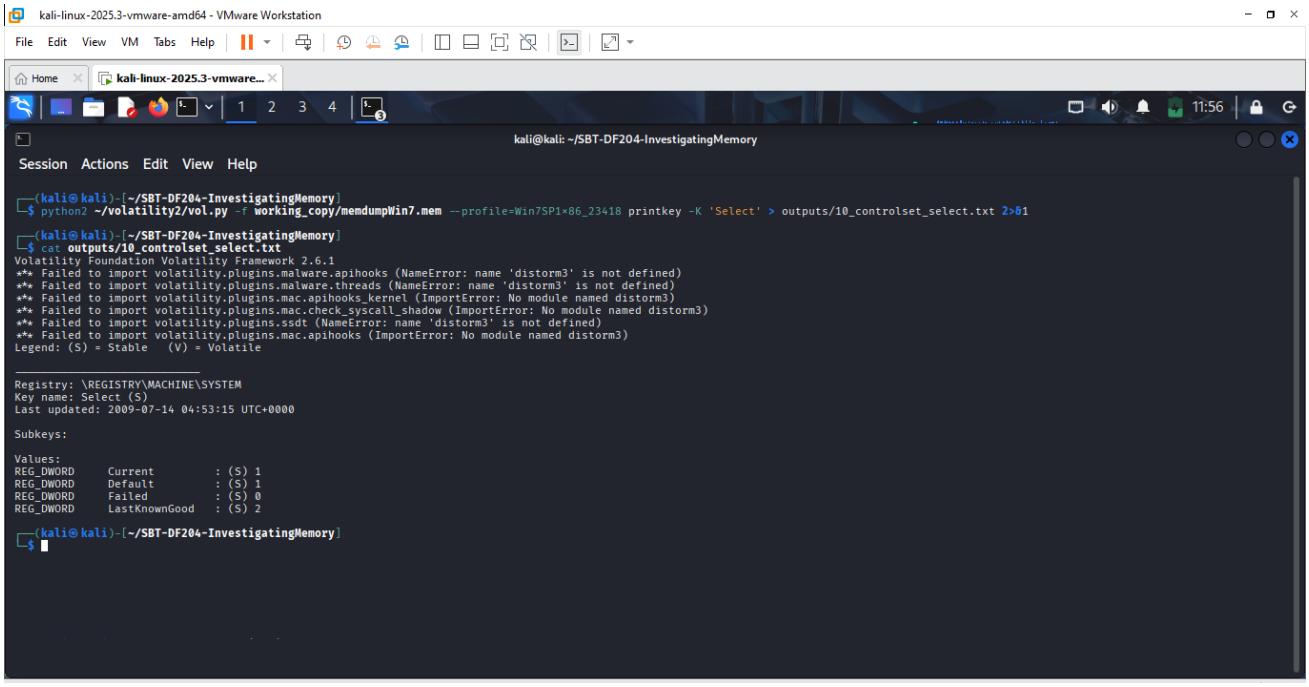


Fig 2.9 SYSTEM Registry ControlSet Identification.

Explanation

I examined the SYSTEM registry's Select key to identify which Windows ControlSet was active in the captured system. The Current value was 1, indicating that ControlSet001 should be used when examining the relevant device and system configuration records.

Registry Findings

Registry Information	Result
Registry Hive	\REGISTRY\MACHINE\SYSTEM
Registry Key	Select
Current	1
Default	1
Failed	0
LastKnownGood	2
Active ControlSet	ControlSet001
Volatility Profile	Win7SP1x86_23418

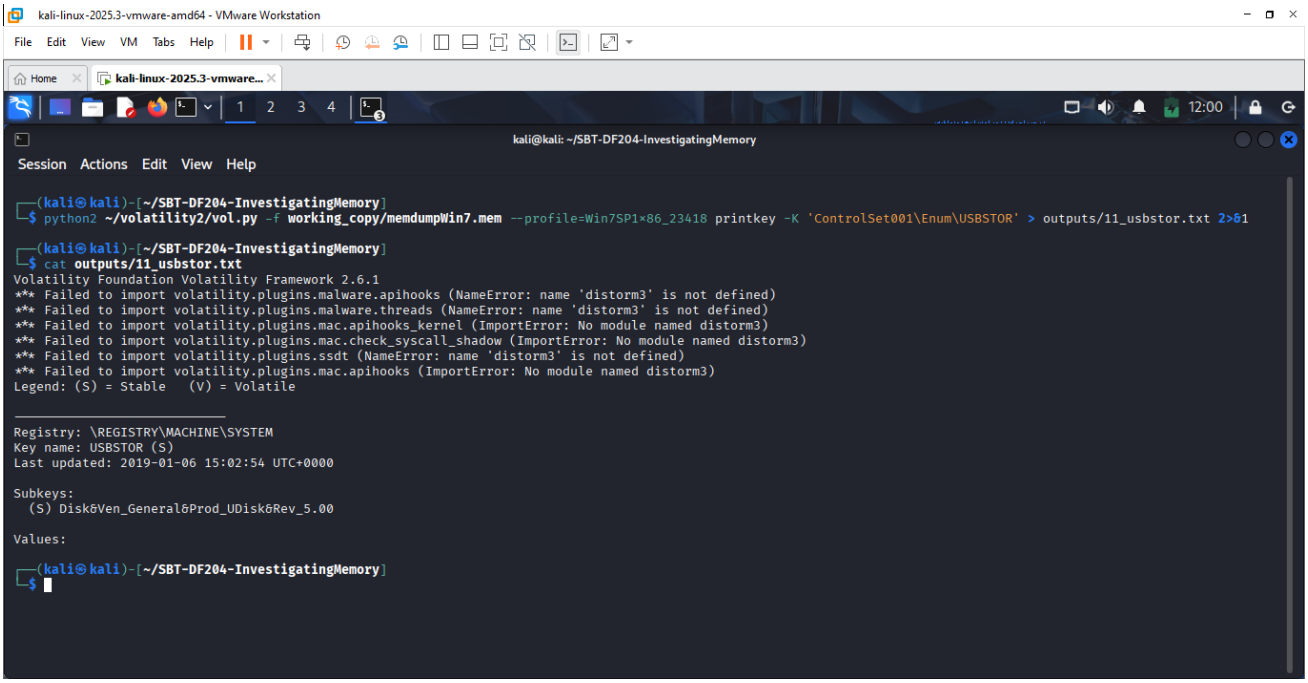


Fig 3.0 USBSTOR Device Identification.

Explanation

The examination of ControlSet001\Enum\USBSTOR identified a registered storage device described as Disk&Ven_General&Prod_UDisk&Rev_5.00. This established the presence of a General UDisk storage record in the Windows registry, although the entry alone did not identify its assigned drive letter.

Evidence Log — E10

E10 — USB Storage Device Identification

Field	Description
Evidence ID	E10
Source Record	SYSTEM registry — ControlSet001\Enum\USBSTOR
Finding	Identified a USB storage device subkey named Disk&Ven_General&Prod_UDisk&Rev_5.00. The USBSTOR key was last updated on 2019-01-06 at 15:02:54 UTC.
Why It Matters	Establishes a recorded USB storage device that may be relevant to the recovered file-copy command. Device identity and drive-letter association require further verification.
Command / Screenshot Reference	Volatility printkey; Fig 3.0 USBSTOR Device Identification.

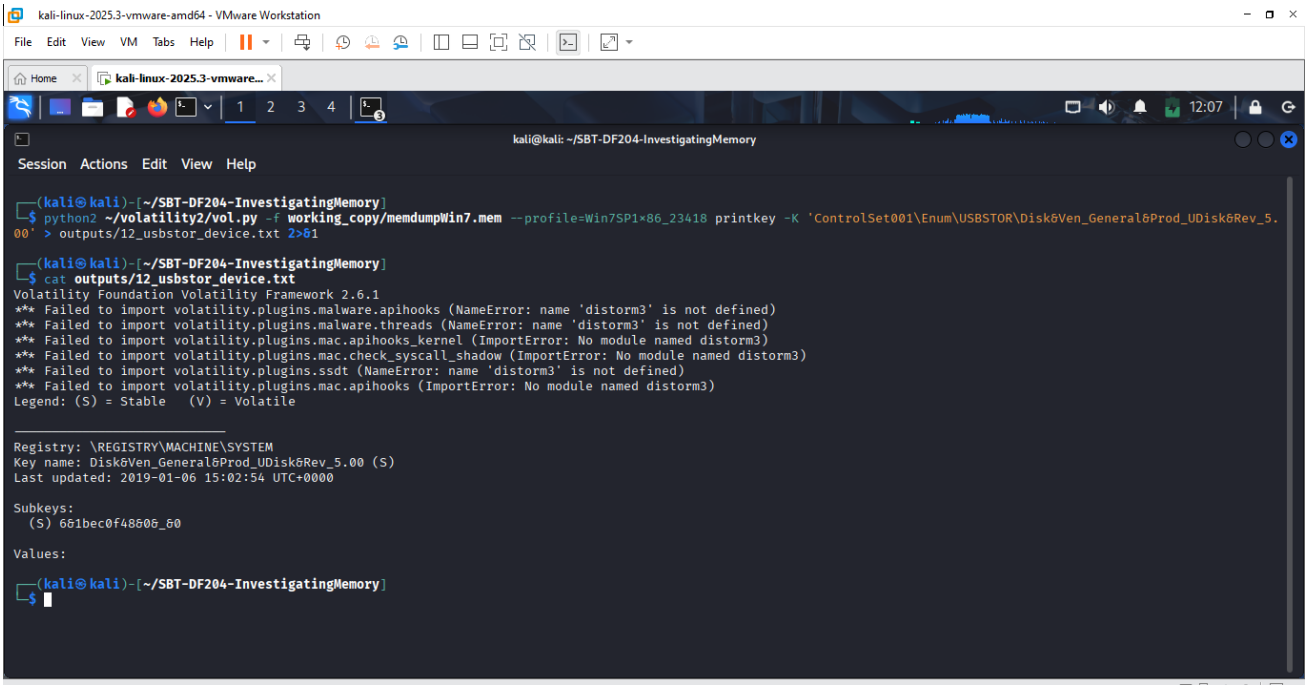


Fig 3.1 USBSTOR Device Instance Identification.

Explanation

I examined the General UDisk registry entry further and recovered a device-instance subkey beneath the USBSTOR record. This provided an additional device identifier that could be compared with the storage and drive-mapping information recovered during subsequent examinations.

Evidence Log — E11

Field	Description
Evidence ID	E11
Source Record	ControlSet001\Enum\USBSTOR\Disk&Ven_General&Prod_UDisk&Rev_5.00
Finding	Identified a device-instance subkey associated with the General UDisk storage device.
Why It Matters	Provides an identifier for correlating USB storage evidence with drive-letter assignments and other registry artefacts.
Screenshot Reference	Fig 3.1 USBSTOR Device Instance Identification.

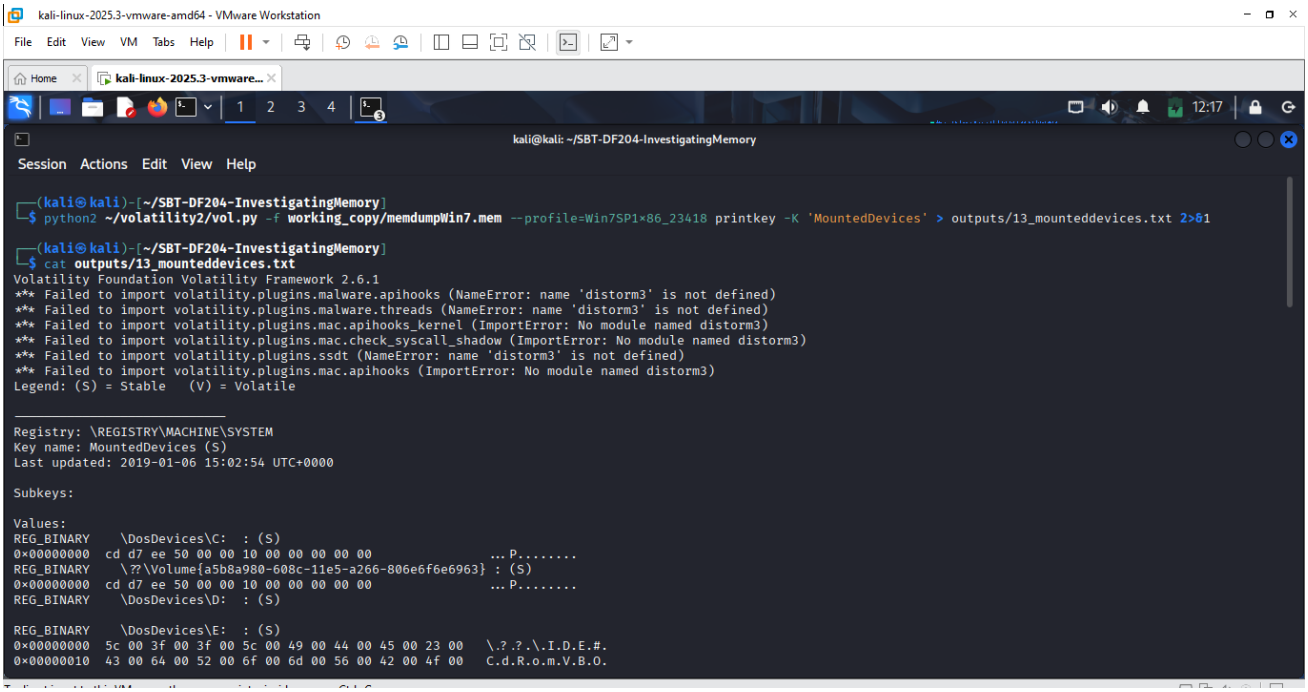


Fig 3.2 MountedDevices Drive Letter Investigation.

Explanation

The MountedDevices examination displayed Windows registry values associated with drive letters and mounted volumes. These entries provided the starting point for determining how Windows recorded storage devices and whether the destination drive referenced in the recovered command could be identified.

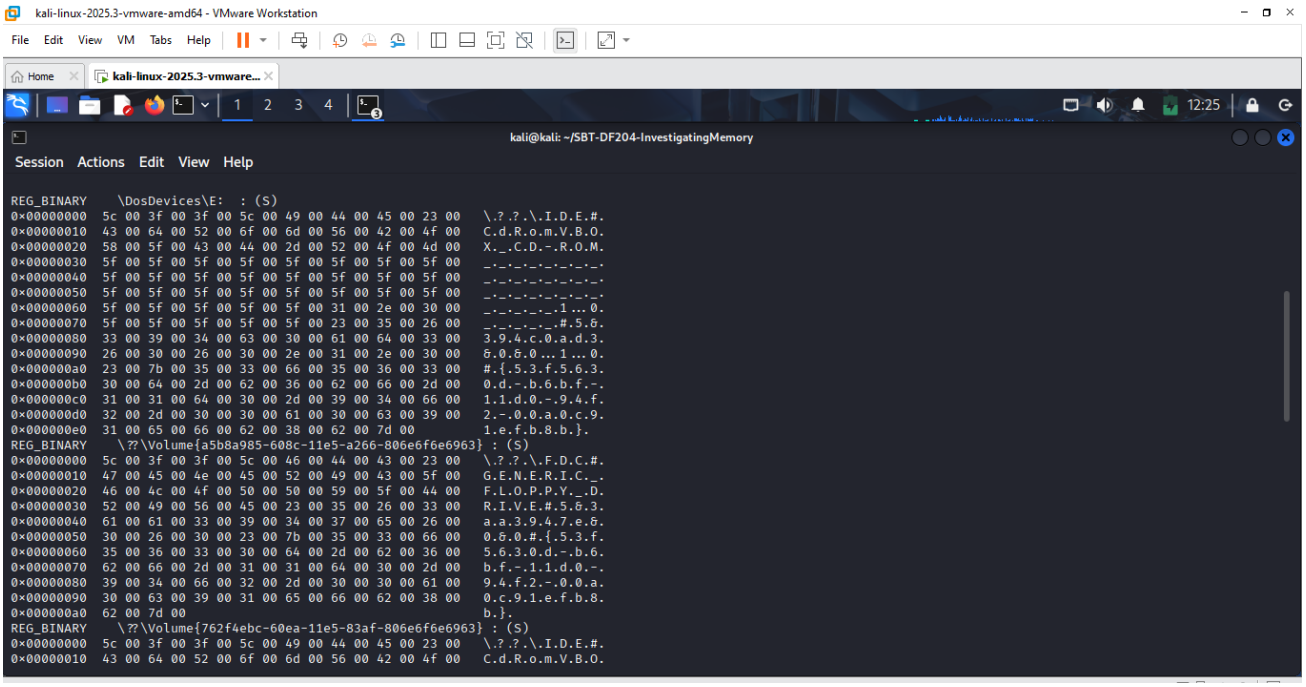


Fig 3.3 MountedDevices Drive Letter Investigation page 1.

Explanation

The continuation displayed additional mounted-volume identifiers and binary device-path information. I examined these records to distinguish ordinary drive entries from storage-device mappings relevant to the suspected file-copy operation.

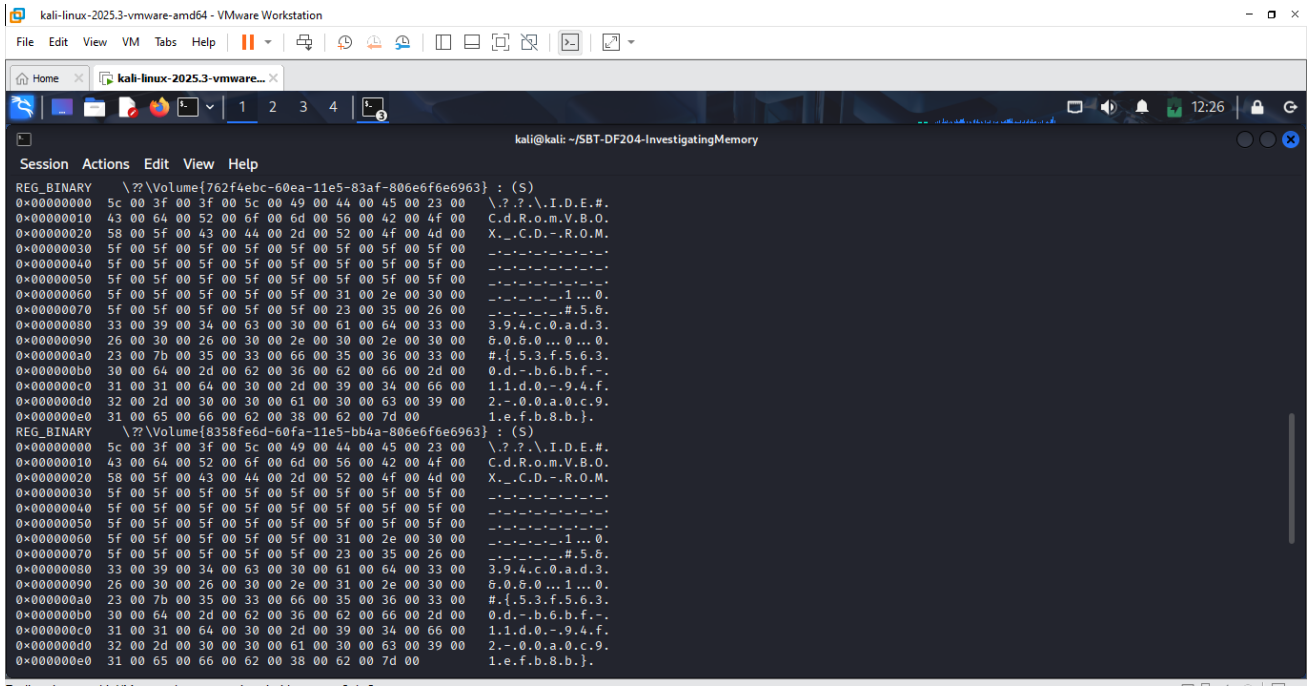


Fig 3.4 MountedDevices Drive Letter Investigation page 2.

Explanation

This section of the output contained further volume records and their associated binary data. Comparing the available device-path information was necessary because a drive-letter entry by itself does not identify the

physical device associated with that letter.

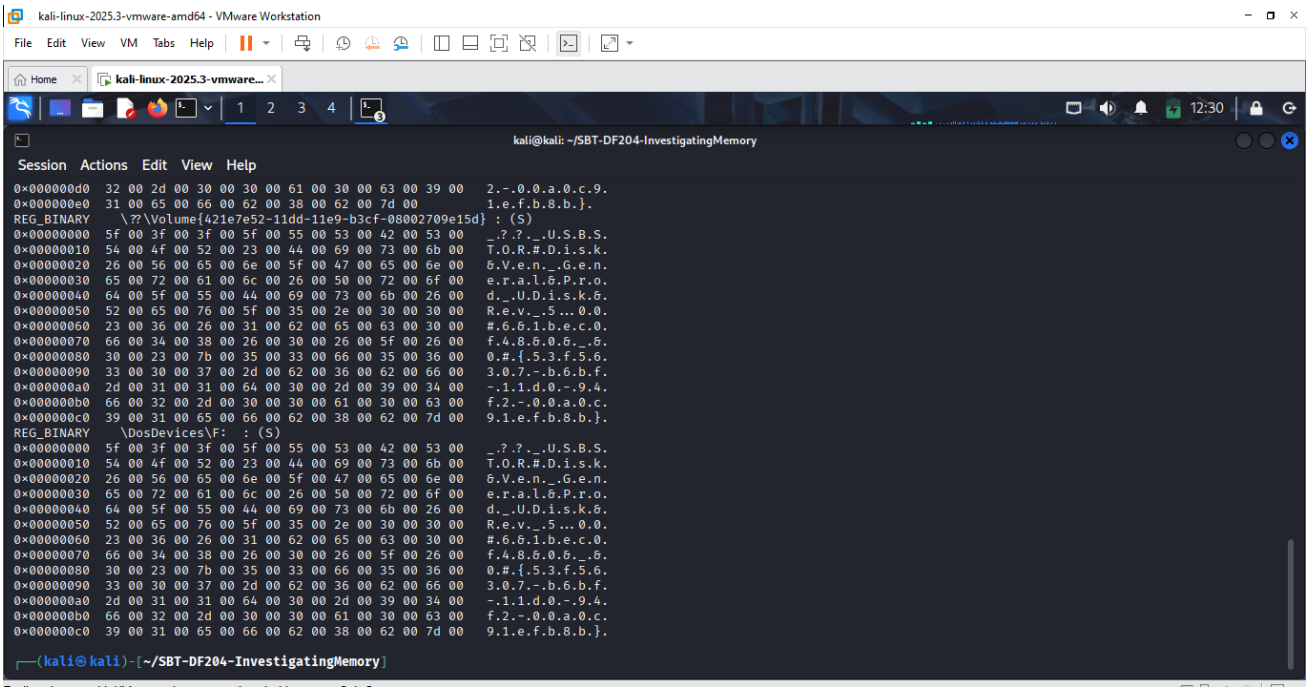


Fig 3.5 MountedDevices Drive Letter Investigation page 3.

Explanation

The investigation identified the \DosDevices\F: registry value containing a device path associated with the General UDisk USB storage device. This established an important connection between the destination drive recorded in the copy command and the removable-storage evidence recovered from the registry.

EVIDENCE LOG — E12

E12 — Drive F: and USB Storage Mapping

Field	Description
Evidence ID	E12
Source Record	SYSTEM registry — MountedDevices
Finding	The \DosDevices\F: registry value contains USBSTOR device-path information identifying General UDisk. A corresponding volume record contains matching device information.
Why It Matters	Establishes a recorded relationship between drive F: and the USB storage device, corroborating the destination referenced in the recovered file-copy command.
Screenshot Reference	MountedDevices Drive Letter Investigation page

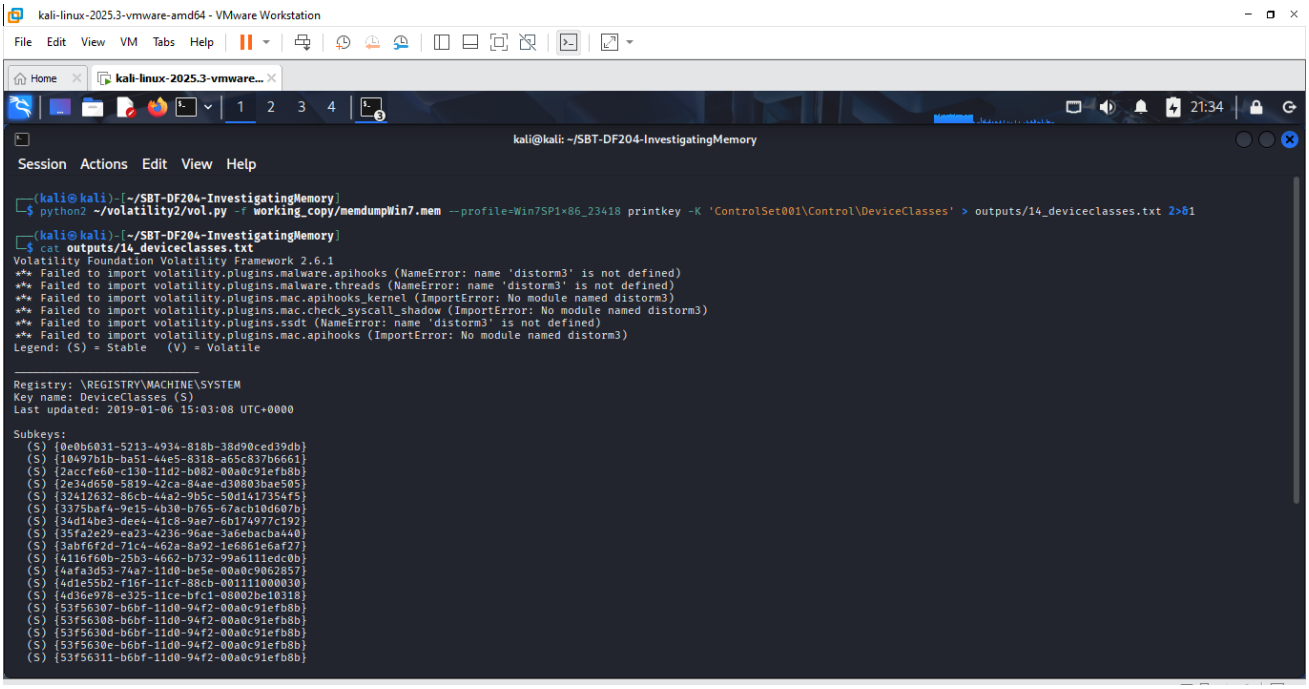


Fig 3.6 Device Classes Registry Investigation.

Explanation

I examined the Windows DeviceClasses registry location to identify device-interface records available in memory. The results contained several interface identifiers, including entries associated with disk and volume devices that required further examination.

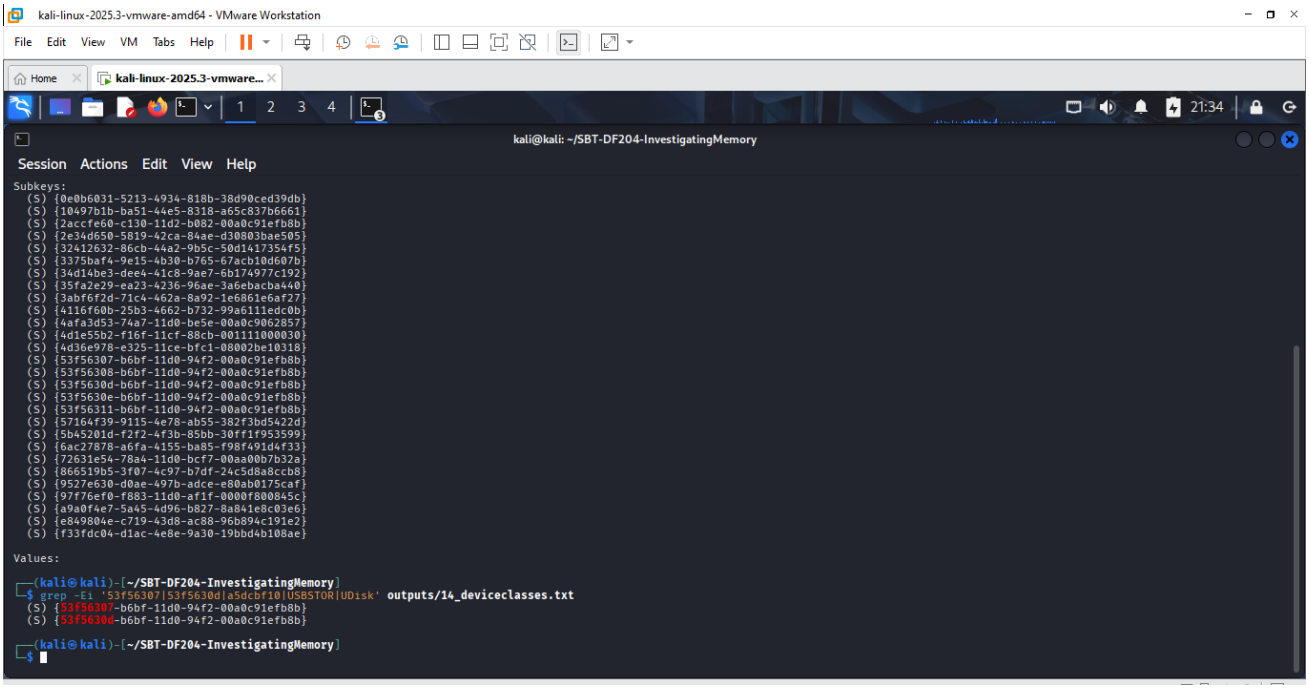


Fig 3.7 Device Classes Registry Investigation page 1.

Explanation

The remaining output displayed additional device-interface class identifiers recovered from the SYSTEM registry. The examination identified disk and volume interface classes, providing specific registry locations for

investigating the General UDisk device.

Evidence Log — E13

E13 — DeviceClasses Registry Examination

Field	Description
Evidence ID	E13
Source Record	ControlSet001\Control\DeviceClasses
Finding	Identified disk and volume device-interface class GUIDs in the SYSTEM registry.
Why It Matters	Provides registry locations for further examination of device-interface records and possible correlation with the General UDisk USB storage device.
Screenshot Reference	Fig 3.6 Device Classes Registry Investigation page

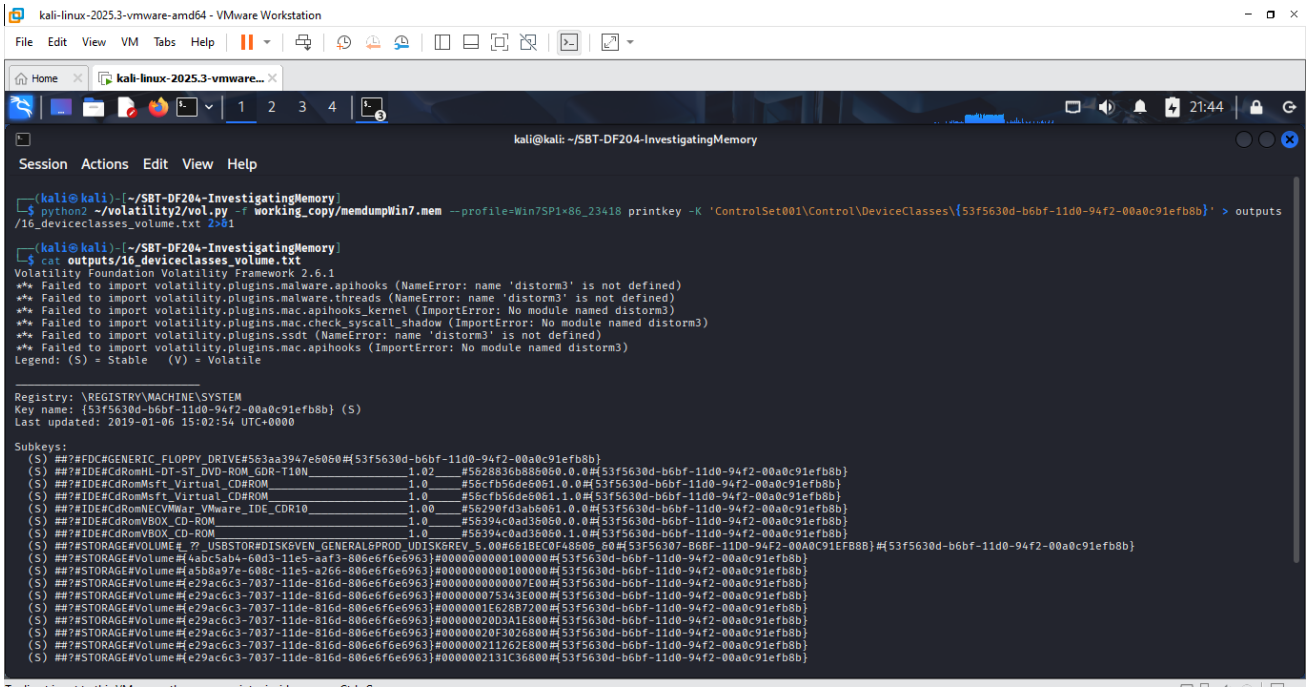


Fig 3.8 Device Classes Volume Interface.

Explanation

I examined the volume-interface registry class and identified an entry containing the General UDisk USBSTOR device path. This supported the earlier findings by showing that Windows also recorded the device within its volume-interface information.

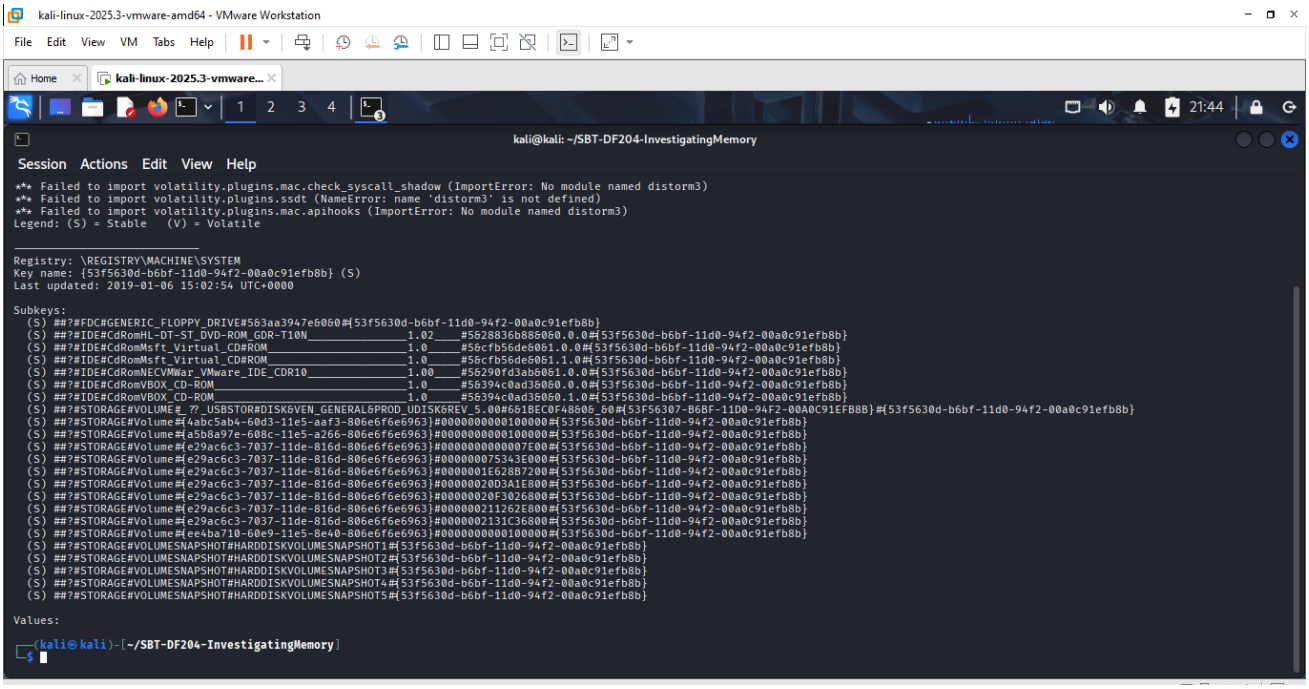


Fig 3.9 Device Classes Volume Interface page 1.

Explanation

The continuation displayed further volume-interface records, including the entry associated with General UDisk. This provided additional registry evidence supporting the storage-device identification, although it did not establish exactly when the device was connected.

Evidence Log — E14

E14 — DeviceClasses Volume Interface Investigation

Field	Description
Evidence ID	E14
Source Record	ControlSet001\Control\DeviceClasses\{53f5630d-b6bf-11d0-94f2-00a0c91efb8b}
Finding	A volume-interface registry subkey contains a USBSTOR device path identifying General UDisk, revision 5.00.
Why It Matters	Corroborates the USB storage and volume-registration evidence recovered during earlier examinations.
Screenshot Reference	Fig 3.9 Device Classes Volume Interface.

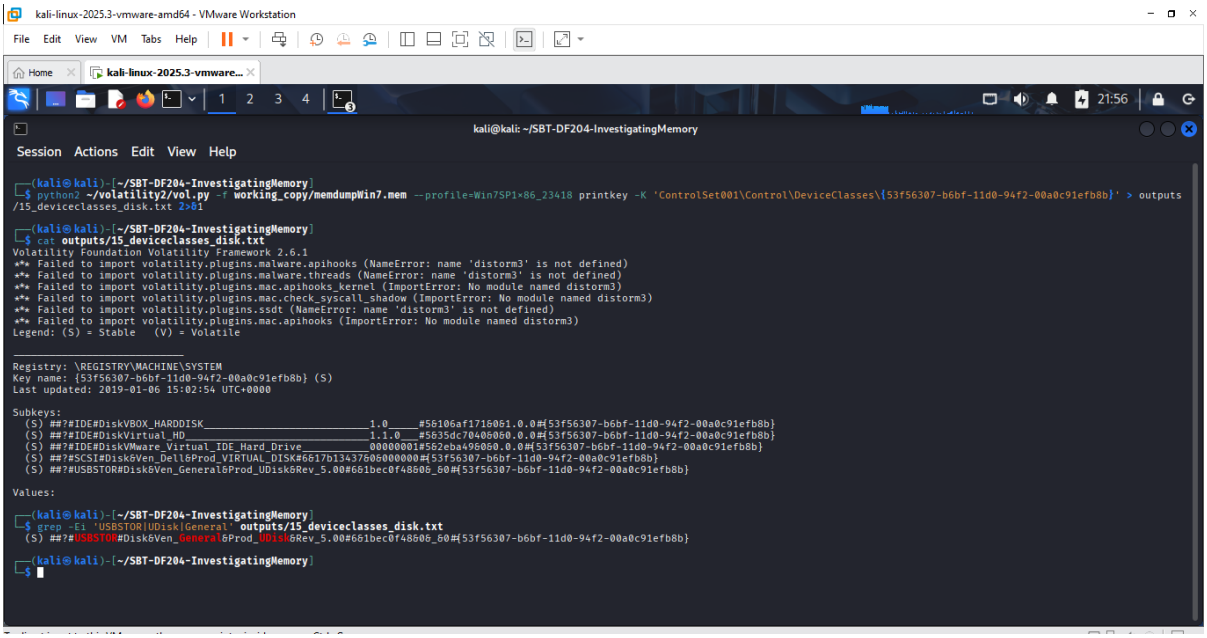


Fig 4.0 Device Classes Disk Interface.

Explanation

The disk-interface examination identified a registry entry referencing the General UDisk USB storage device. This finding was consistent with the earlier USBSTOR and volume-interface results, strengthening the relationship between the recovered device records.

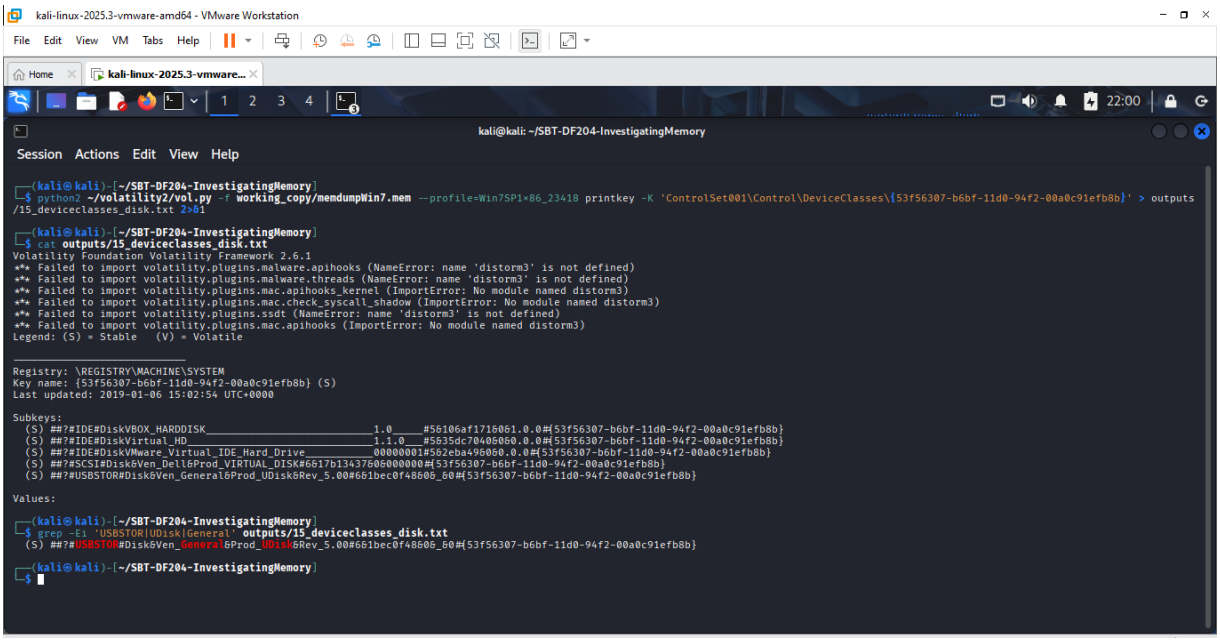


Fig 4.1 Device Classes Disk Interface.

Explanation

The displayed search results confirmed that the disk-interface records contained a USBSTOR path referencing General UDisk, revision 5.00. This supplied another matching device identifier that could be compared with the drive-letter evidence recovered from MountedDevices.

Evidence Log — E15

E15 — DeviceClasses Disk Interface Examination

Field	Description
Evidence ID	E15
Source Record	ControlSet001\Control\DeviceClasses\{53f56307-b6bf-11d0-94f2-00a0c91efb8b}
Finding	The disk-interface registry key contains a USBSTOR device path referencing General UDisk, revision 5.00.
Why It Matters	Corroborates the USB storage device's registration and provides additional evidence supporting the relationship between the device and previously recovered drive-mapping records.
Command / Screenshot Reference	Volatility printkey; Fig 4.1 Device Classes Disk Interface.

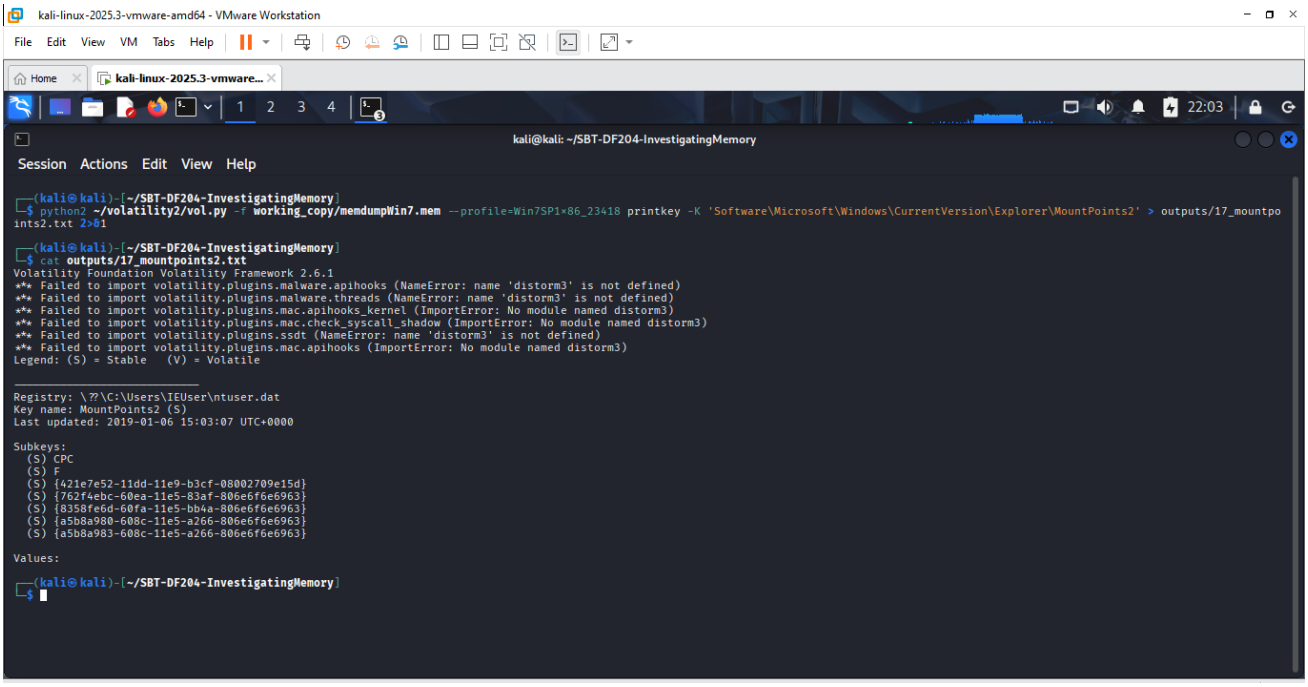


Fig 4.2 IEUser MountPoints2 Registry Investigation.

Explanation

I examined the MountPoints2 registry records recovered from the IEUser profile and identified a subkey named F alongside several volume identifiers. This showed that the Windows profile contained a record associated with the same drive letter referenced in the recovered file-copy command.

Evidence Log — E16

E16 — IEUser MountPoints2 Investigation

Field	Description
Evidence ID	E16
Source Record	Volatility printkey; C:\Users\IEUser\ntuser.dat\Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2
Finding	The IEUser registry hive contains a MountPoints2 subkey named F and several volume GUID entries. The parent key was last updated on 6 January 2019 at 15:03:07 UTC.
Why It Matters	Establishes a user-profile association with drive F:, supporting further correlation with the recovered copy command and USB evidence. It does not identify the person operating the workstation.
Screenshot Reference	Fig 4.1 IEUser MountPoints2 Registry Investigation.

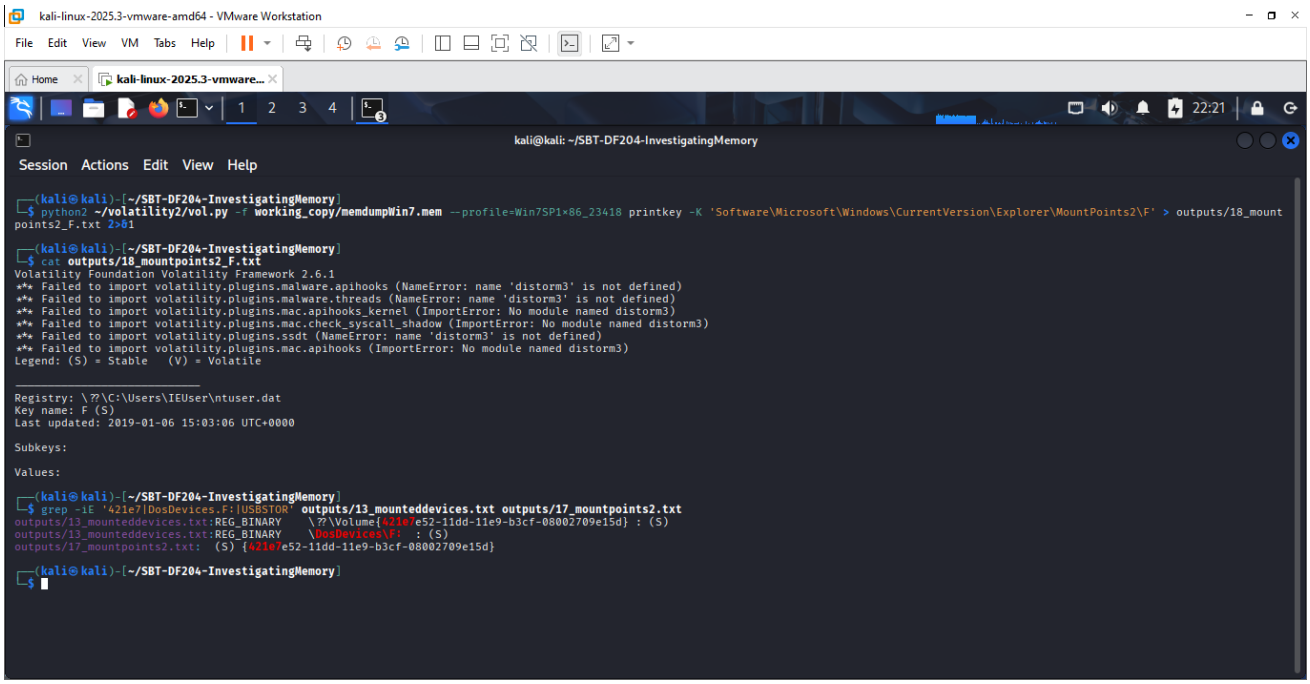


Fig 4.3 MountPoints2 Drive F Verification.

Explanation

The examination of the F subkey showed a last-updated timestamp of 6 January 2019 at 15:03:06 UTC. Comparing the saved registry outputs also identified a matching volume GUID, providing additional support for the relationship between the IEUser profile and the mounted-volume records.

Evidence Log — E17

Field	Description
Evidence ID	E17
Source Record	IEUser NTUSER.DAT, MountPoints2\F; SYSTEM MountedDevices
Finding	The IEUser profile contains a drive F: record last updated at 15:03:06 UTC. The saved MountedDevices and MountPoints2 outputs contain a matching volume GUID.
Why It Matters	Corroborates the relationship between the user-profile records and the mounted-volume evidence.
Screenshot Reference	Fig 4.3 MountPoints2 Drive F Verification.

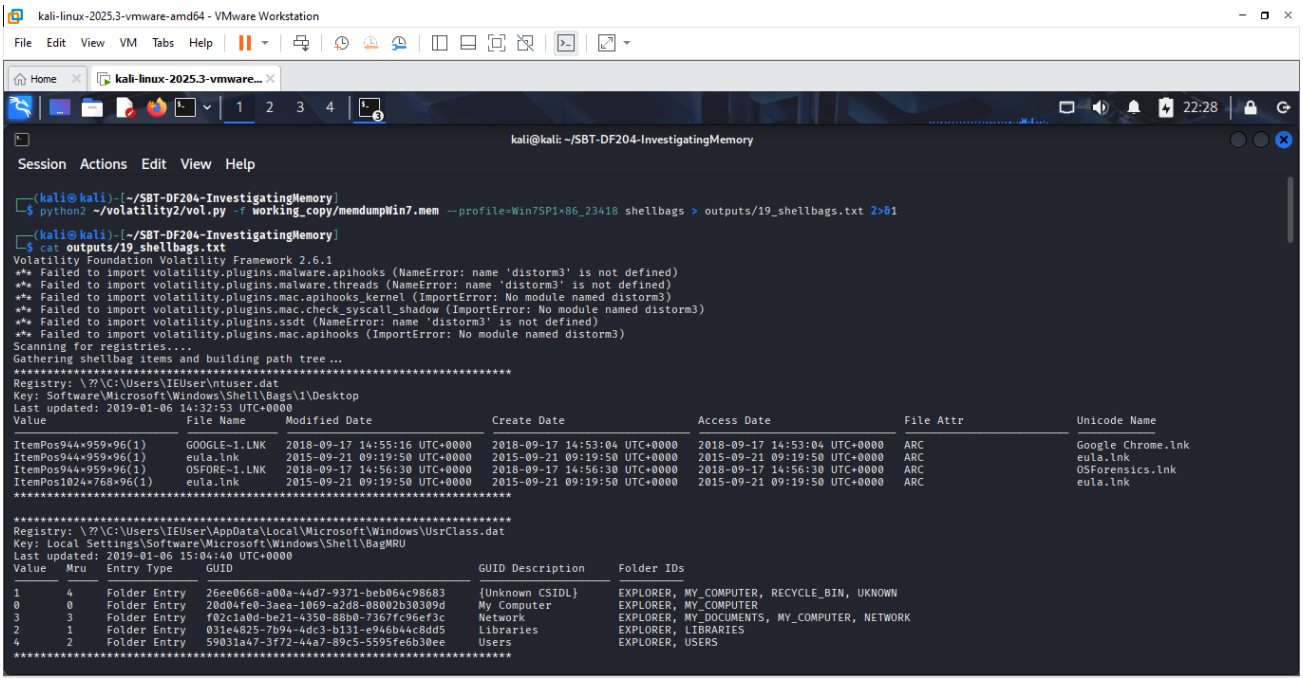


Fig 4.4 ShellBags Investigation.

Explanation

I used the Volatility shellbags plugin to examine folder-history information preserved in the IEUser registry files. The output included records from NTUSER.DAT and UsrClass.dat, providing information about folders and File Explorer-related settings recorded by Windows.

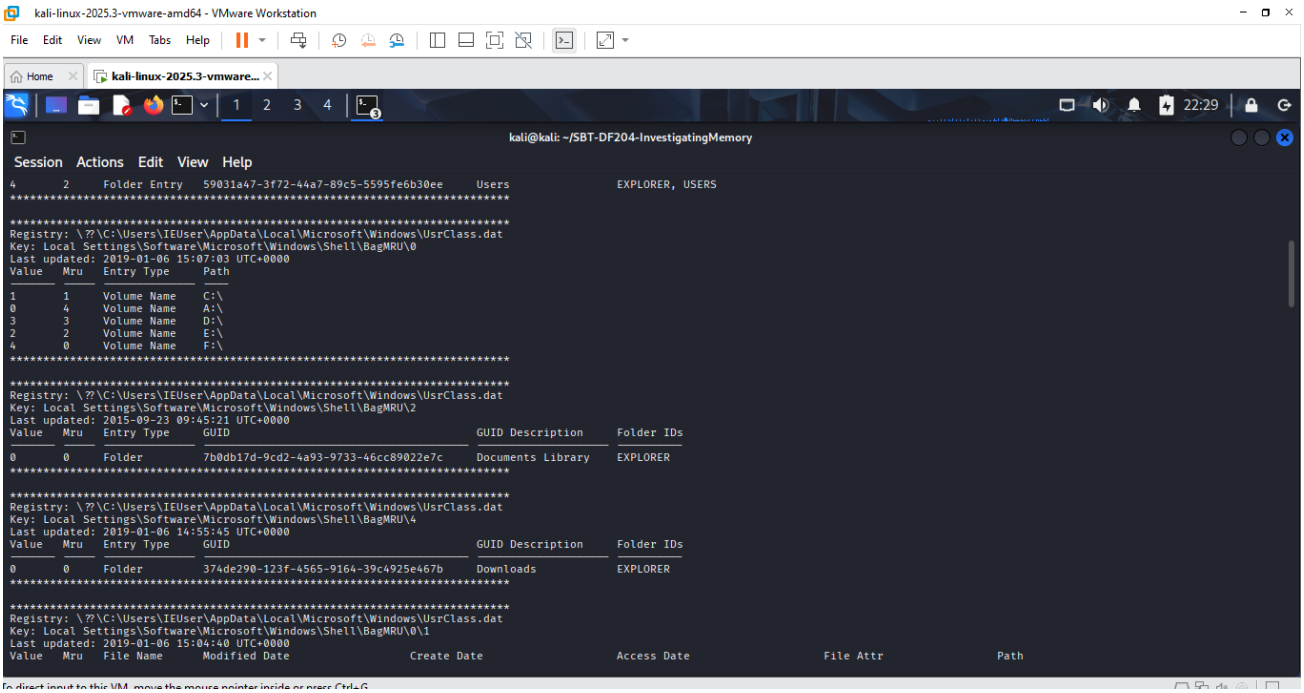


Fig 4.5 ShellBags Investigation page 1.

Explanation

The recovered ShellBag information contained entries for several drive letters, including F:, and a folder entry identifying Downloads. These records were relevant because the previously recovered command history also referenced the Downloads directory and destination drive F:.

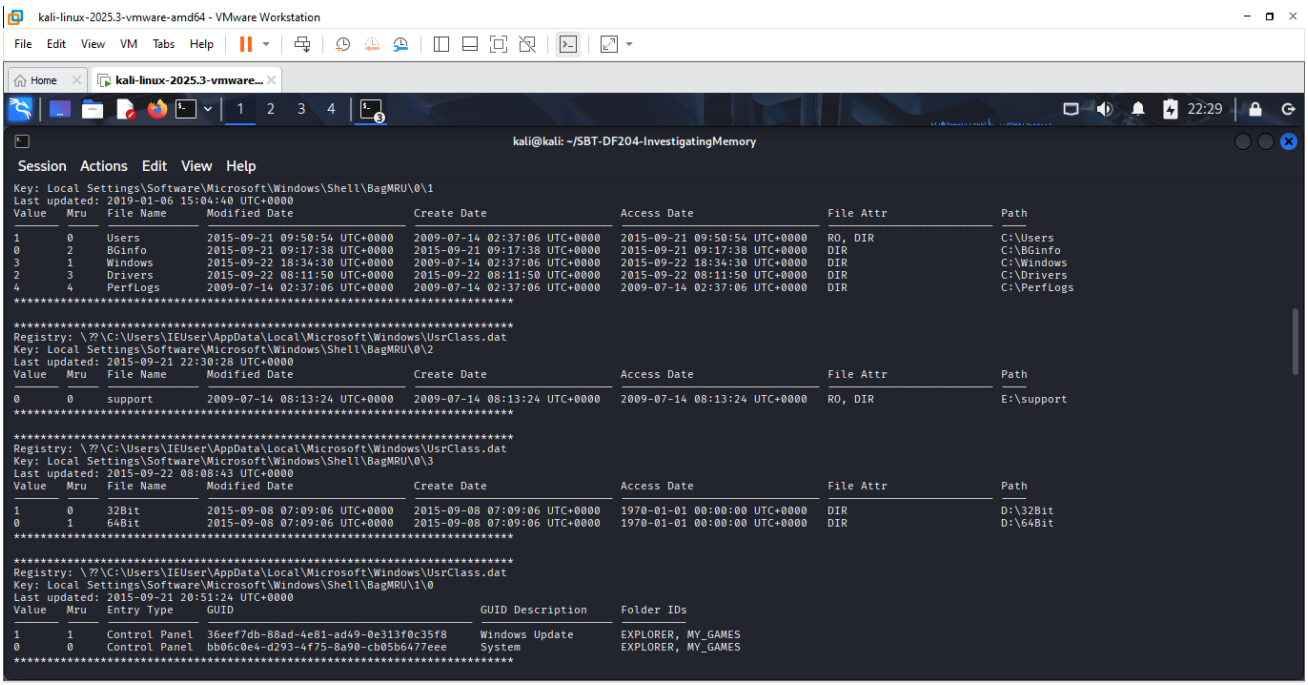


Fig 4.6 ShellBags Investigation page 2.

Explanation

The investigation recovered additional folder paths, including Windows system directories and an E:\support

folder. These entries helped establish the range of folder-history information available in memory, although they did not independently identify the document involved in the suspected transfer.

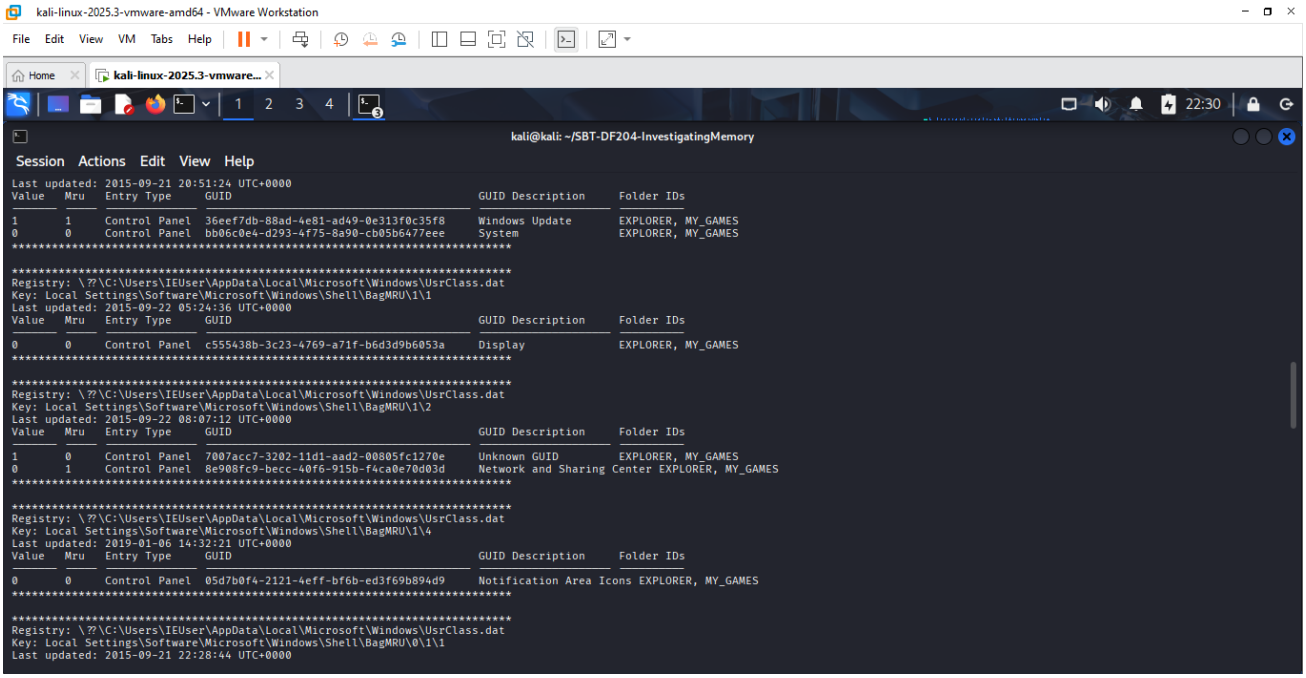


Fig 4.7 ShellBags Investigation page 3.

Explanation

The ShellBag output displayed further Windows Explorer-related entries, including Control Panel and system configuration locations. These records demonstrated that the recovered folder-history information covered different areas of the Windows environment rather than only the folders relevant to the suspected transfer.

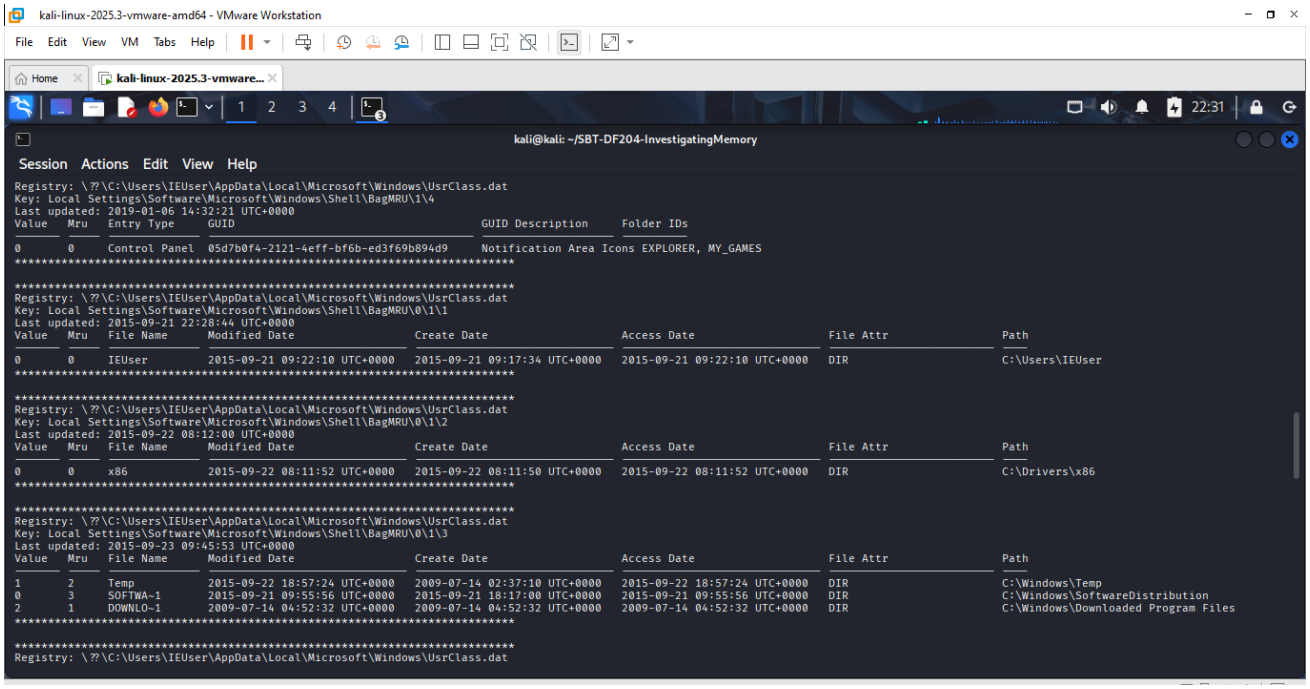


Fig 4.8 ShellBags Investigation page 4.

Explanation

This section contained additional folder entries associated with the IEUser profile and Windows directories. The recovered paths provided more context about the folder-history records stored in the user profile, but they did not establish any particular file-copy operation.

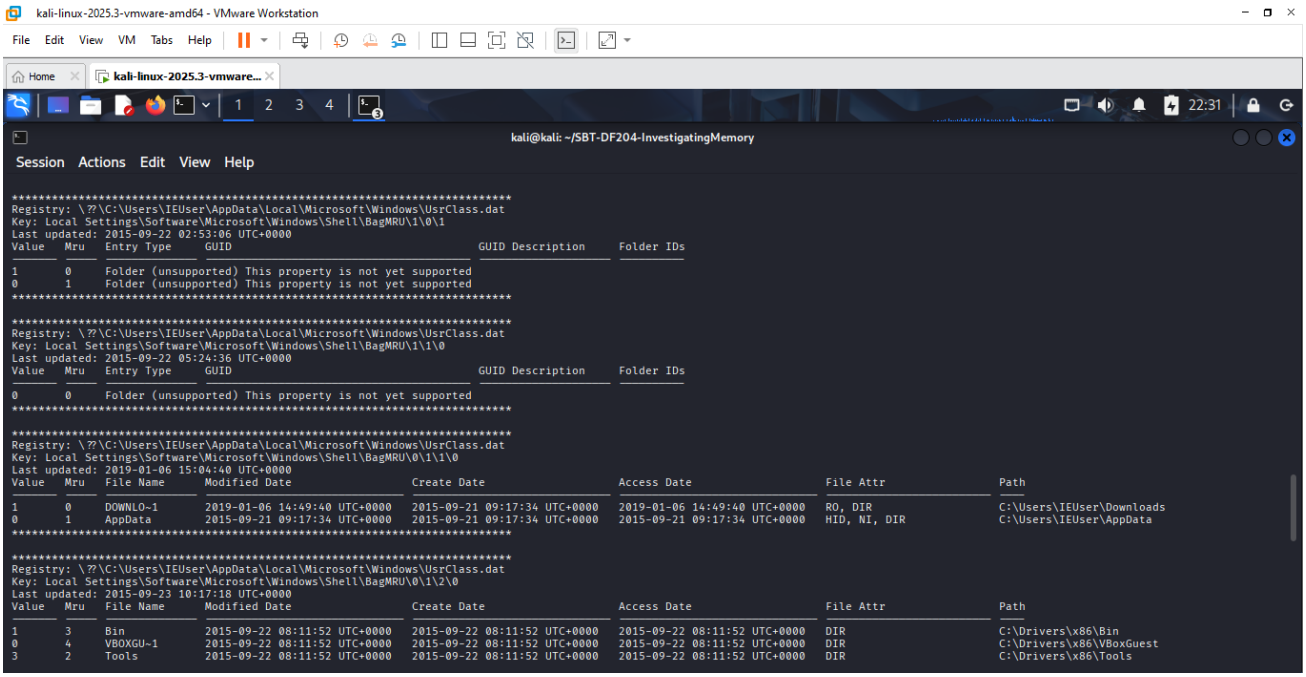


Fig 4.9 ShellBags Investigation page 5.

Explanation

The examination displayed ShellBag entries relating to the Downloads folder and other user-profile locations. The Downloads record was particularly relevant because it corresponded to the source directory identified in the recovered Command Prompt activity.

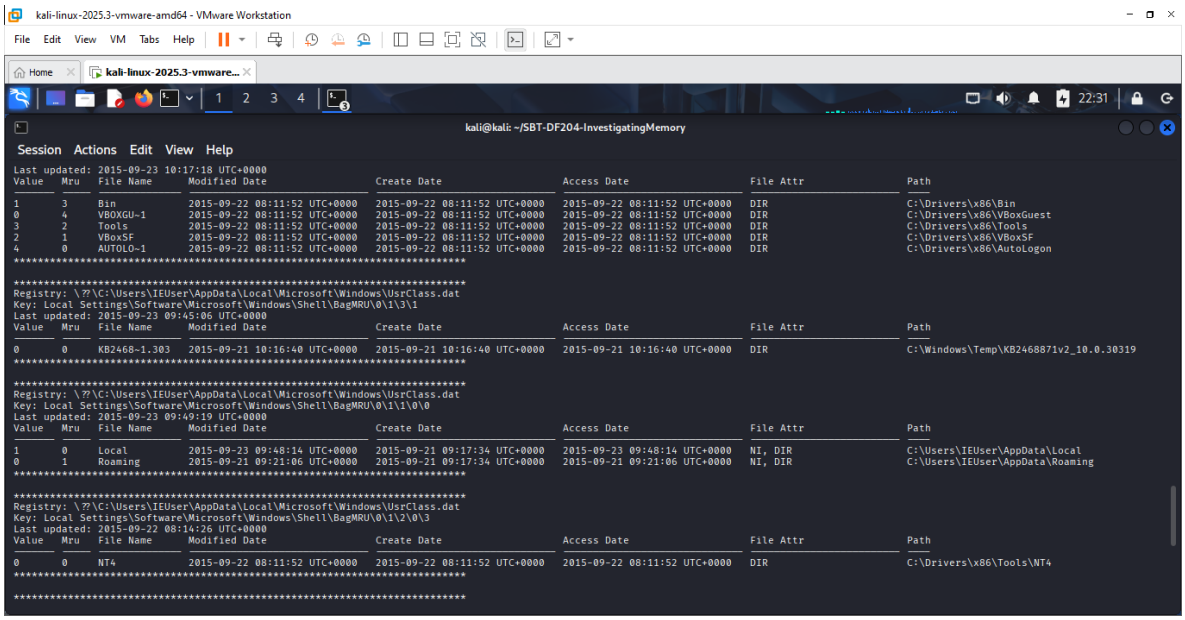


Fig 5.0 ShellBags Investigation page 6.

Explanation

Further ShellBag records identified directories associated with the IEUser account, including application-data and temporary-file locations. These entries added context to the profile's folder history but were not direct evidence that secret_file.docx had been copied.

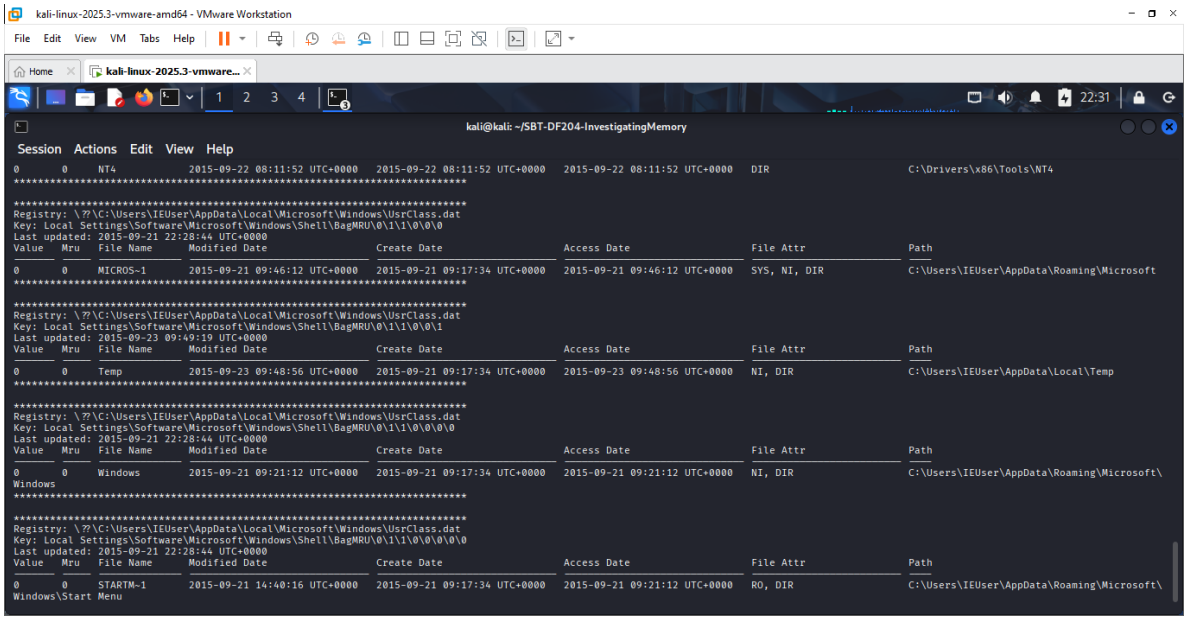


Fig 5.1 ShellBags Investigation page 7.

Explanation

The remaining records included locations within the IEUser application-data directories and Windows Start Menu structure. Their presence showed additional Explorer-related information available in memory, although these entries did not directly establish the suspected transfer.

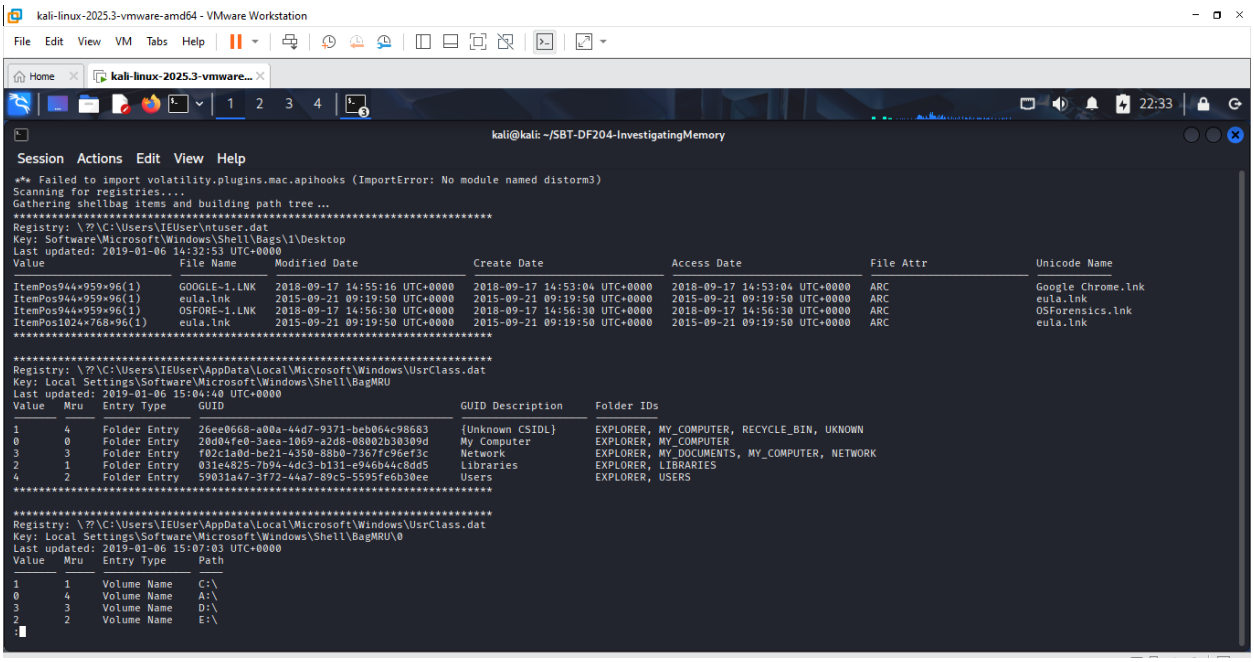


Fig 5.2 ShellBags Investigation page 8.

Explanation

This screenshot displayed further ShellBag output, including desktop-related records and entries for several drive letters. The recovered volume references were useful for comparing the profile's folder-history information with the mounted-drive evidence already identified.

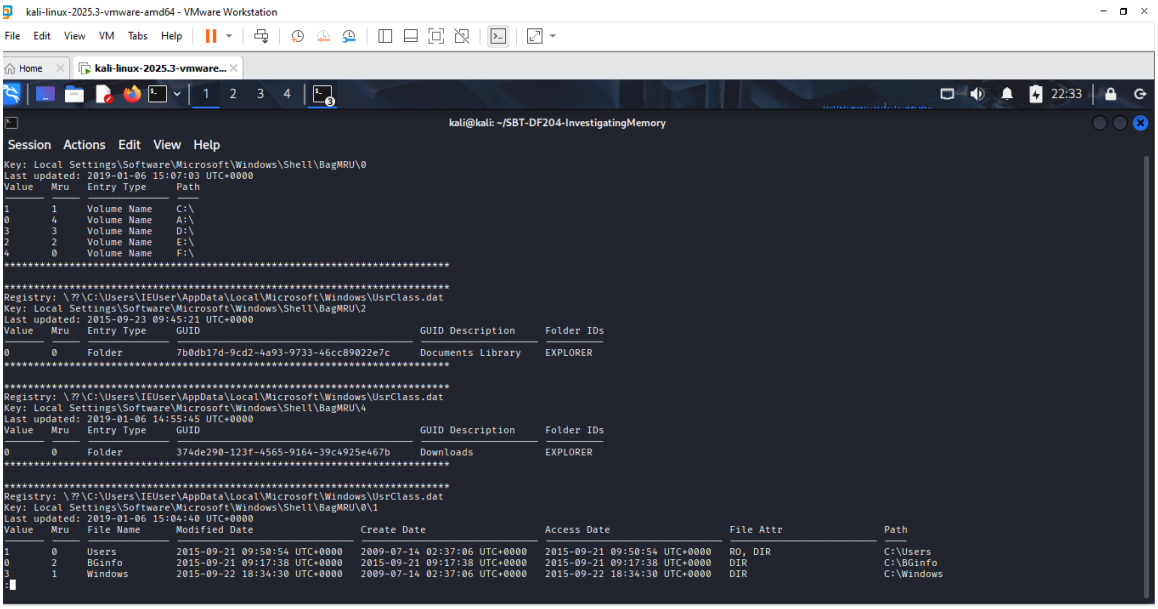


Fig 5.3 ShellBags Investigation page 9.

Explanation

The continuation included folder-history records relating to Windows directories, the Downloads location, and other recorded volumes. These results helped confirm that the examined user profile contained filesystem-navigation information relevant to the investigation.

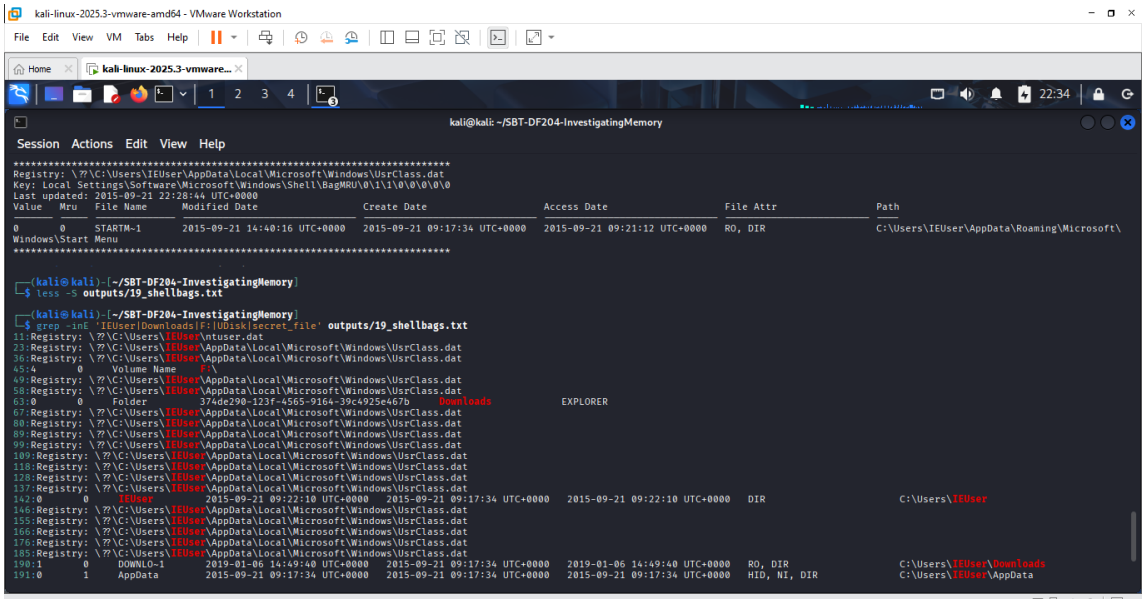


Fig 5.4 ShellBags Investigation page 10.

Explanation

I searched the saved ShellBag output for references to IEUser, Downloads, and other relevant artefacts. The search highlighted entries associated with the examined user profile and helped locate the Downloads-related records within the larger output.

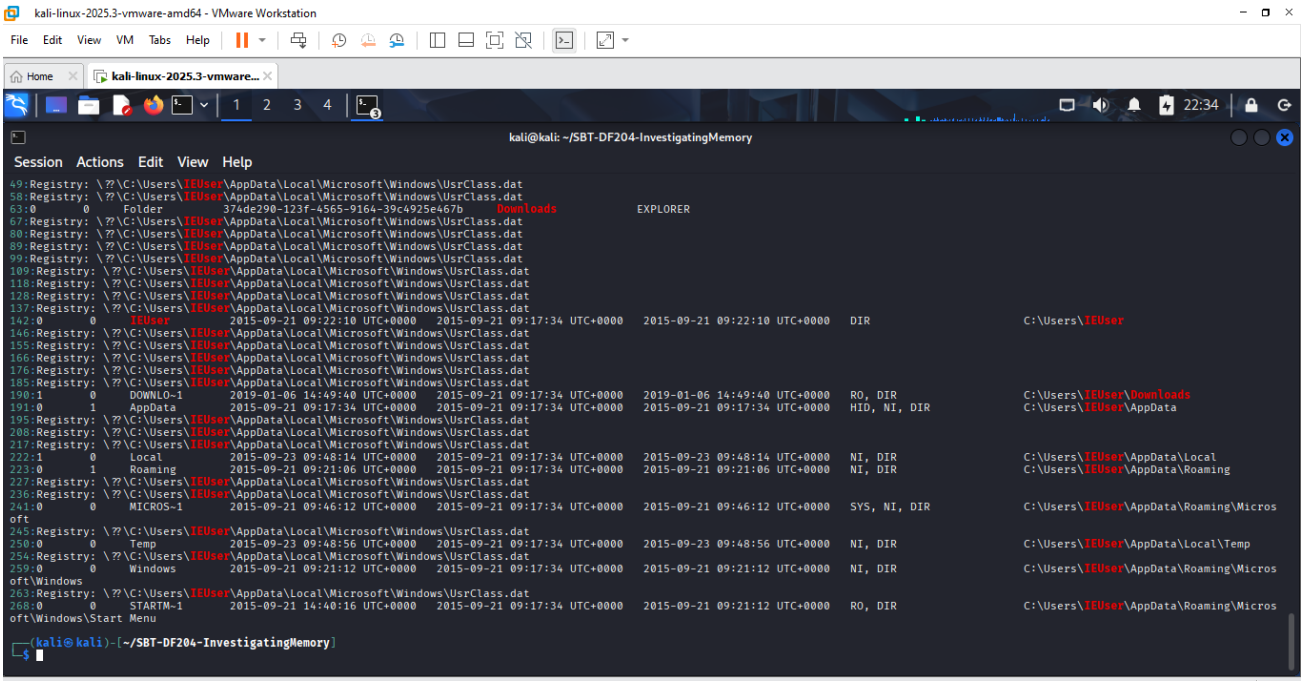


Fig 5.5 ShellBags Investigation page 11.

Explanation

The remaining search results displayed additional IEUser-related registry and folder paths, including the Downloads directory. These findings supported the presence of relevant folder-history artefacts but did not prove that the document was copied at the time recorded in any particular ShellBag entry.

EVIDENCE LOG — E18

E18 — ShellBag File Explorer Investigation

Field	Description
Evidence ID	E18
Source Record	Volatility 2.6.1, shellbags, profile Win7SP1x86_23418; IEUser's NTUSER.DAT and UsrClass.dat
Finding	Identified ShellBag records referencing C:\Users\IEUser\Downloads and volume F:\. The Downloads folder record contains modified and accessed timestamps of 2019-01-06 14:49:40 UTC. The registry key listing drive F: was last updated at 15:07:03 UTC.

Field	Description
Why It Matters	Corroborates the presence of Downloads and drive F: in user-profile File Explorer artefacts. These records support the reconstructed activity context but do not independently prove a file transfer or identify the person responsible.
Command / Screenshot Reference	outputs/19_shellbags.txt; 5 ShellBags Investigation page 1- 11.

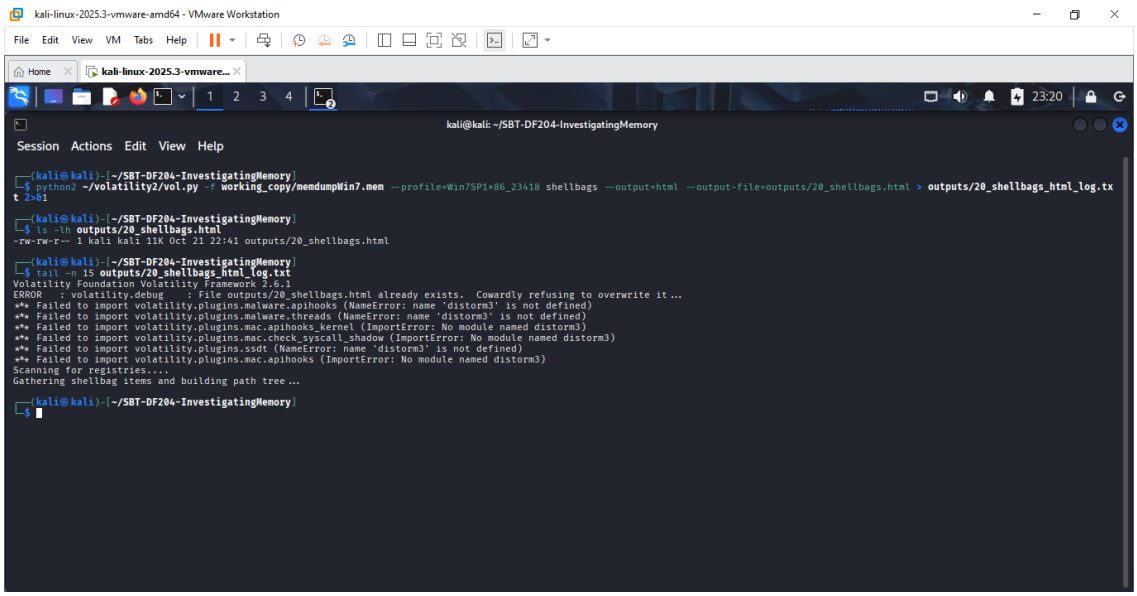


Fig 5.6 ShellBags Investigation page 12.

Explanation

I attempted to preserve the recovered ShellBag information in HTML format using Volatility. The output showed that an HTML file with the selected name already existed, so Volatility refused to overwrite it; this did not affect the previously recovered ShellBag text results.

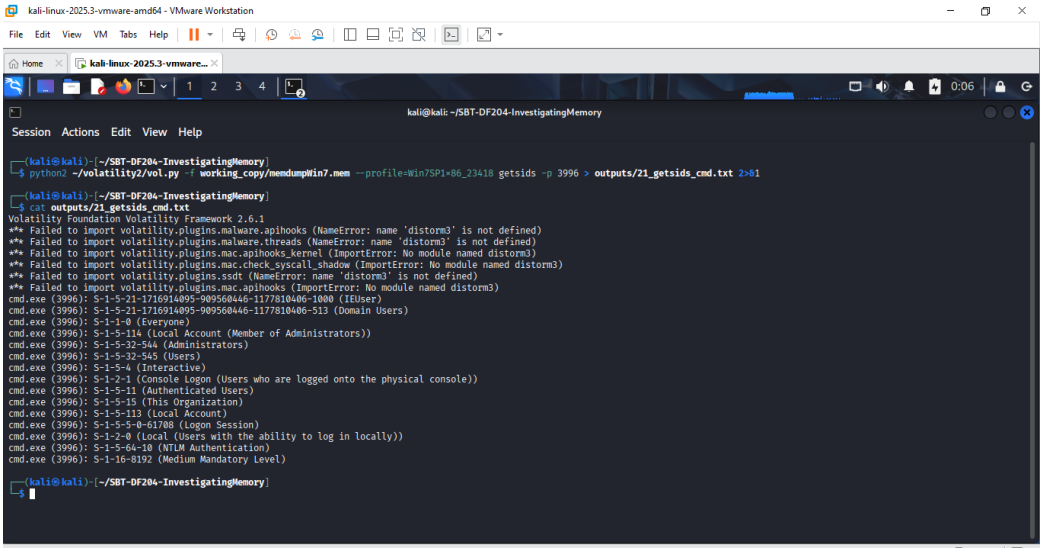


Fig 5.7 Command Prompt User SID Investigation.

Explanation
I examined the security identifiers associated with Command Prompt, PID 3996, using the getsids plugin. The results identified the IEUser Windows account, with a security identifier ending in -1000, establishing the account context associated with the recovered command history.

E19 — Command Prompt User Account Investigation

Field	Description
Evidence ID	E19
Source Record	Volatility 2.6.1 — getsids, PID 3996
Finding	Command Prompt was associated with the IEUser account, SID ending in -1000. The process token also contained interactive-logon and administrator-group identifiers.
Why It Matters	Associates the Command Prompt used for the recovered copy command with the IEUser Windows account. It does not establish the identity of the person who entered the command.
Screenshot Reference	Fig 5.7 Command Prompt User SID Investigation.

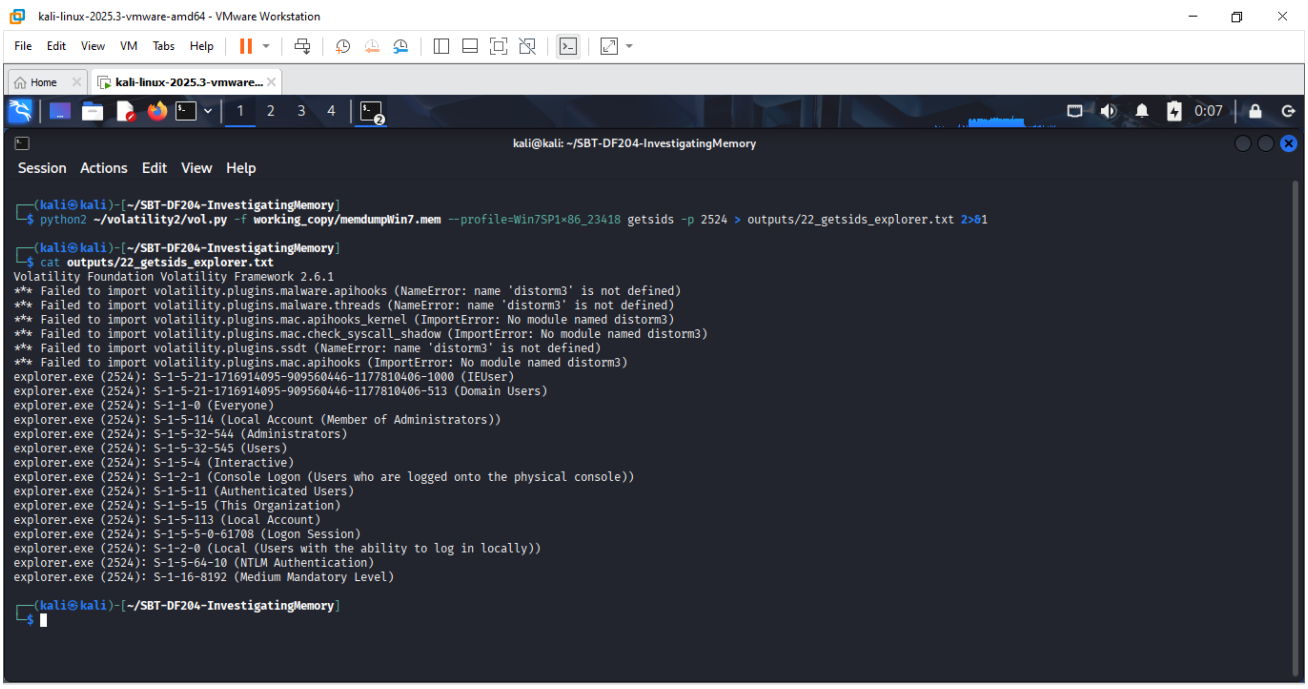


Fig 5.8 Windows Explorer User SID Investigation.

Explanation
The getsids examination of Windows Explorer, PID 2524, returned the same IEUser account security identifier recovered from Command Prompt. This established that both processes operated under the same Windows account context, although it did not identify the individual using the computer.

E20 — Windows Explorer User Account Investigation

Field	Description
Evidence ID	E20
Source Record	Volatility 2.6.1 — getsids, PID 2524
Finding	Windows Explorer was associated with the same IEUser account SID recovered from Command Prompt.
Why It Matters	Corroborates the shared Windows account context of Windows Explorer and Command Prompt.
Screenshot Reference	Fig 5.8 Windows Explorer User SID Investigation.

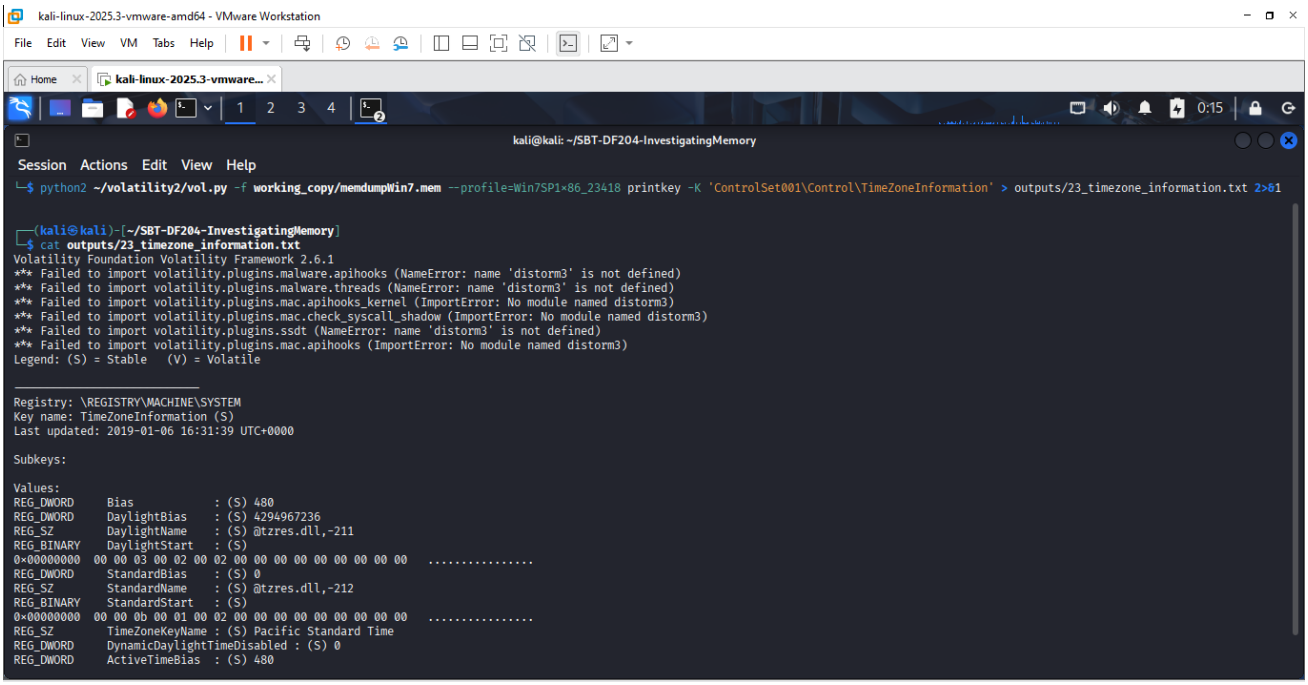


Fig 5.9 Timestamp Consistency and TimeZone Check.

Explanation

I examined the Windows TimeZoneInformation registry key and identified the configured time zone as Pacific Standard Time, with an offset of eight hours behind UTC. This information was useful for checking the relationship between the image's local timestamp and the UTC time reported by Volatility.

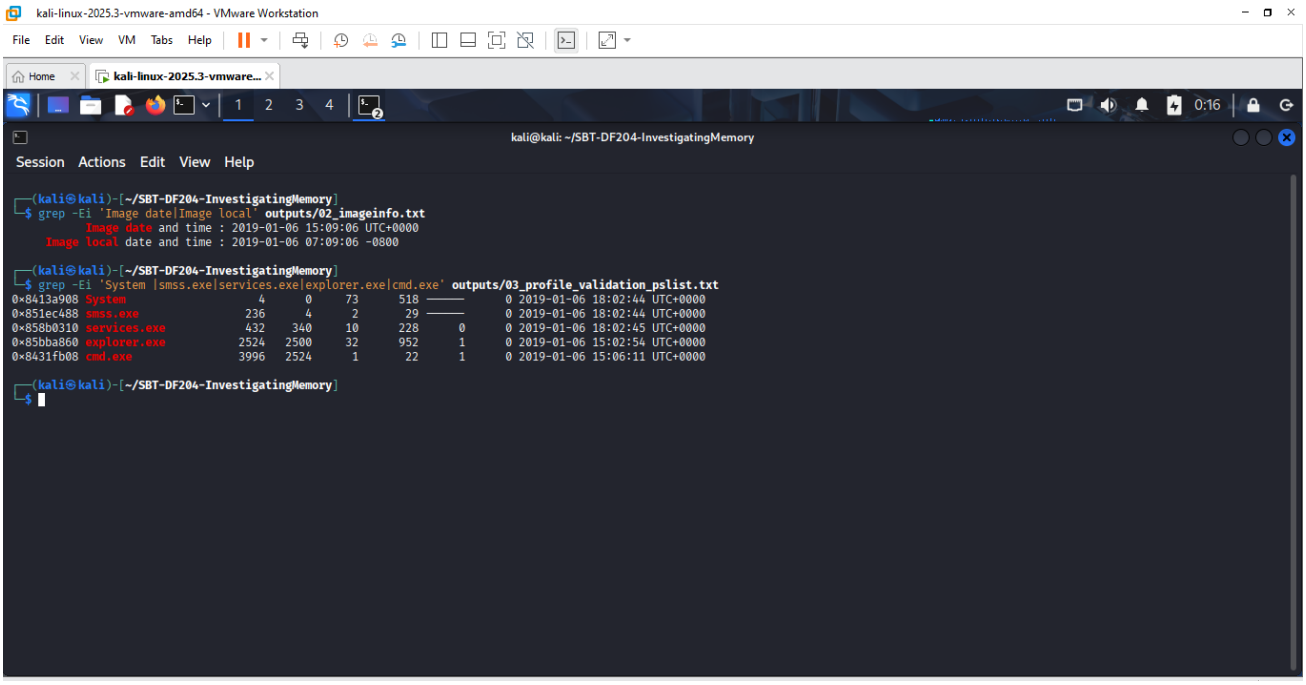


Fig 6.0 Timestamp Consistency and TimeZone Check page 1.

Explanation

The comparison showed that the memory image had a reported timestamp of 15:09:06 UTC, while some system processes had creation timestamps around 18:02 UTC on the same date. This inconsistency could not be resolved from the recovered evidence and was therefore treated as a limitation when interpreting the investigation timeline.

EVIDENCE LOG — E21

E21 — Time-Zone Configuration and Timestamp Consistency

Field	Description
Evidence ID	E21
Source Record	Volatility printkey, imageinfo, and pslist
Finding	The Windows system was configured for Pacific Standard Time (UTC−08:00). However, several process creation timestamps and the TimeZoneInformation registry last-write timestamp occur after the reported memory-image time.
Why It Matters	Establishes the time-zone configuration while identifying unresolved timestamp inconsistencies that must be considered when reconstructing events.
Screenshot Reference	Fig 5.9 Timestamp Consistency and TimeZone Check.

Preliminary UTC Timeline

The following events are supported by the evidence recovered during our investigation.

UTC Date and Time	Evidence ID	Observed Event	Interpretation and Limitation
2019-01-06 14:49:40	E18	ShellBag record for C:\Users\IEUser\Downloads contains this modified/accessed timestamp.	Folder metadata; not proof that a file was copied at this time.
2019-01-06 15:02:54	E03	Windows Explorer, PID 2524, recorded as starting.	Establishes the reported process creation time.
2019-01-06 15:02:54	E10	General UDisk USBSTOR registry key last updated.	Supports device registration; not necessarily the precise insertion time.
2019-01-06 15:03:06	E17	IEUser's MountPoints2\F registry key last updated.	Establishes a recorded profile association with F:.
2019-01-06 15:03:07	E16	IEUser's MountPoints2 parent key last updated.	Provides additional mounted-volume context.
2019-01-06 15:04:30	E03	Chrome process, PID 2388, recorded as starting.	Establishes browser-process activity, not a confirmed download.
2019-01-06 15:04:40	E18	Downloads-related ShellBag registry key last updated.	Supports folder-history activity but does not establish a precise folder visit or file-copy time.
2019-01-06 15:06:11	E03, E06	Command Prompt, PID 3996, recorded as starting under Explorer.	Identifies the process associated with recovered command history.
2019-01-06 15:07:03	E18	ShellBag registry key containing the F: volume entry last updated.	Supports File Explorer-related volume history; not proof of copying at this time.
2019-01-06 15:07:15	E03	FTK Imager, PID 2596, recorded as starting.	May relate to evidence acquisition; its precise role is unverified.
2019-01-06 15:08:32	E07	Chrome-associated UDP endpoint recorded.	Establishes network activity, not a confirmed file transfer.
2019-01-06 15:09:06	E02	Volatility reports the memory-image UTC timestamp.	Provides the reported image time, subject to the identified inconsistencies.

Limitations

1. Examination of Memory Evidence Only

The investigation was based on the supplied Windows memory image rather than a complete forensic examination of the computer's storage devices. Because memory evidence represents information available in RAM at the time of acquisition, some activities may not have been preserved, while other records may have been incomplete or overwritten.

2. No Independent Examination of the Destination File

The strongest evidence of the suspected transfer came from the recovered command copy secret_file.docx F: and the console message 1 file(s) copied.. Although this indicates that Windows reported a successful copy operation, the investigation did not independently recover the destination file or verify its contents on the USB storage device.

3. Limited Personal Attribution

The recovered security identifiers associated Command Prompt and Windows Explorer with the IEUser account. However, account-level evidence cannot establish who was physically operating the workstation, and the available material did not rule out another person having access to the same account.

4. Uncertainty About Authorization

The investigation established evidence of file-copy activity, but the recovered memory records did not contain sufficient information to determine whether the action was authorized. Therefore, the file-copy operation could not automatically be classified as illegal or unauthorized.

5. Timestamp Inconsistencies

Some recovered process creation timestamps occurred after the memory image's reported capture time of 15:09:06 UTC. The TimeZoneInformation registry also contained a later last-updated timestamp, creating uncertainty about the reliability of parts of the chronological record.

Although the system's Pacific Standard Time configuration explained the difference between local and UTC image timestamps, it did not resolve these additional inconsistencies. As a result, the investigation preserved the timestamps as recovered and avoided assigning unsupported times to events.

6. Limitations of Command and Registry Artefacts

The recovered command history did not provide an exact execution timestamp for the file-copy command. Similarly, USBSTOR, MountedDevices, MountPoints2, and DeviceClasses records supported the identification of the storage device and drive mapping but did not independently establish the precise connection time or the identity of the person who used the device.

7. Limitations of Network and ShellBag Evidence

The network examination identified browser-related activity, closed HTTP connections, and a listening SSH service, but these findings did not demonstrate that a document was transferred over the network.

ShellBag records provided useful folder and volume information, including references to Downloads and drive F:. However, ShellBag timestamps could not be treated as direct evidence of file modification or copying.

8. Tool Compatibility Limitations

During the investigation, Volatility displayed warnings relating to the unavailable distorm3 dependency. Although the plugins required for the completed examination produced usable results, the missing dependency prevented some additional plugins from loading.

Conclusion

This investigation examined a Windows memory image to determine whether the available digital evidence supported a suspected unauthorized file transfer. Using Volatility 2.6.1, I examined the computer's processes, network connections, command history, Windows registry records, removable-storage information, File Explorer artefacts, and account security identifiers.

The most significant finding was the recovered Command Prompt activity showing that secret_file.docx was copied from the IEUser Downloads directory to drive F:. The corresponding console output displayed 1 file(s) copied., providing evidence that Windows reported successful completion of the operation.

Further examination of USBSTOR, MountedDevices, DeviceClasses, and MountPoints2 identified a General UDisk USB storage device and established a recorded association between that device and drive F:. The ShellBag results also contained references to the Downloads directory and the destination drive, providing additional context for the activity.

The process and security-identifier examinations established that the relevant Command Prompt and Windows Explorer processes were associated with the IEUser Windows account. However, these findings could not identify the person who entered the command or establish whether the activity was authorized.

The investigation therefore reached a high-confidence finding that Windows reported a successful file-copy operation to a drive associated with USB storage. Nevertheless, the absence of an independently recovered destination file, unresolved timestamp inconsistencies, and limitations in personal attribution prevented a definitive conclusion that an unauthorized transfer had occurred.

Overall, the case demonstrates how information recovered from computer memory can be combined to reconstruct important aspects of suspected activity. It also highlights the need to support forensic conclusions with identifiable evidence while recognizing the difference between what digital artefacts reveal and what they cannot establish.

Screenshots

Fig 1.0 Creating the Lab folder.
Fig 1.1 Original Memory Evidence Identification.
Fig 1.2 Original Memory Evidence Identification page 1.
Fig 1.3 SHA256 Evidence Integrity Verification.
Fig 1.4 Memory Imageinfo Profile Identification.
Fig 1.5 Memory Profile Validation Pslist.
Fig 1.6 Memory Profile Validation Pslist page 1.
Fig 1.7 Memory Profile Validation Pslist page 2.
Fig 1.8 Memory Profile Validation Hivelist.
Fig 1.9 PSSCAN Process Investigation.
Fig 2.0 PSSCAN Process Investigation page 1.
Fig 2.1 PSTREE Process Relationships.
Fig 2.2 PSTREE Process Relationships page 1.
Fig 2.3 NETSCAN Network Connections.
Fig 2.4 NETSCAN Network Connections page 1.
Fig 2.5 CMDSCAN Command History.
Fig 2.6 CONSOLES Command Execution Evidence.
Fig 2.7 CONSOLES Command Execution Evidence page 1.
Fig 2.8 CONSOLES Command Execution Evidence page 2.
Fig 2.9 SYSTEM Registry ControlSet Identification.
Fig 3.0 USBSTOR Device Identification.
Fig 3.1 USBSTOR Device Instance Identification.
Fig 3.2 MountedDevices Drive Letter Investigation.
Fig 3.3 MountedDevices Drive Letter Investigation page 1.
Fig 3.4 MountedDevices Drive Letter Investigation page 2.
Fig 3.5 MountedDevices Drive Letter Investigation page 3.
Fig 3.6 Device Classes Registry Investigation.
Fig 3.7 Device Classes Registry Investigation page 1.
Fig 3.8 Device Classes Volume Interface.
Fig 3.9 Device Classes Volume Interface page 1.
Fig 4.0 Device Classes Disk Interface.
Fig 4.1 Device Classes Disk Interface.
Fig 4.2 IEUser MountPoints2 Registry Investigation.
Fig 4.3 MountPoints2 Drive F Verification.
Fig 4.4 ShellBags Investigation.
Fig 4.5 ShellBags Investigation page 1.
Fig 4.6 ShellBags Investigation page 2.
Fig 4.7 ShellBags Investigation page 3.
Fig 4.8 ShellBags Investigation page 4.
Fig 4.9 ShellBags Investigation page 5.
Fig 5.0 ShellBags Investigation page 6.
Fig 5.1 ShellBags Investigation page 7.
Fig 5.2 ShellBags Investigation page 8.
Fig 5.3 ShellBags Investigation page 9.
Fig 5.4 ShellBags Investigation page 10.
Fig 5.5 ShellBags Investigation page 11.

Fig 5.6 ShellBags Investigation page 12.
Fig 5.7 Command Prompt User SID Investigation.
Fig 5.8 Windows Explorer User SID Investigation.
Fig 5.9 Timestamp Consistency and TimeZone Check page 1.
Fig 6.0 Timestamp Consistency and TimeZone Check.

References

ICDFA Materials.